

CHAPTER 1

CLASSIFICATION

In general terms, propulsion is the act of changing the motion of a body with respect to an inertial reference frame. Propulsion systems provide forces that either move bodies initially at rest or change their velocity or that overcome retarding forces when bodies are propelled through a viscous medium. The word *propulsion* comes from the Latin *propulsus*, which is the past participle of the verb *propellere*, meaning “to drive away.” *Jet propulsion* is a type of motion whereby a reaction force is imparted to a vehicle by the momentum of ejected matter.

Rocket propulsion is a class of jet propulsion that produces thrust by ejecting matter, called the working fluid or *propellant*, stored entirely in the flying vehicle. *Duct propulsion* is another class of jet propulsion and it includes turbojets and ram-jets; these engines are more commonly called air-breathing engines. Duct propulsion devices mostly utilize their surrounding medium as the propellant, energized by its combustion with the vehicle’s stored fuel. Combinations of rockets and duct propulsion devices have been attractive for some applications, and one is briefly described in this chapter.

The *energy source* most commonly used in rocket propulsion is *chemical combustion*. Energy can also be supplied by *solar radiation* and by a *nuclear reactor*. Accordingly, the various propulsion devices in use can be divided into *chemical propulsion*, *nuclear propulsion*, and *solar propulsion*. Table 1–1 lists many important propulsion concepts according to their energy source and type of propellant. Radiant energy may originate from sources other than the sun and theoretically includes the transmission of energy by ground-based microwaves and laser beams. Nuclear energy originates in transformations of mass within atomic nuclei and is generated by either fission or fusion. Energy sources are central to rocket performance and several kinds, both within and external to the vehicle, have been investigated. The useful energy

TABLE 1–1. Energy Sources and Propellants for Various Propulsion Concepts

Propulsion Device	Energy Source ^a			Propellant or Working Fluid
	Chemical	Nuclear	Solar	
Turbojet	D/P			Fuel + air
Turbo–ramjet	TFD			Fuel + air
Ramjet (hydrocarbon fuel)	D/P	TFD		Fuel + air
Ramjet (H ₂ cooled)	TFD			Hydrogen + air
Rocket (chemical)	D/P	TFD		Stored propellant
Ducted rocket	TFD			Stored solid fuel + surrounding air
Electric rocket	D/P		D/P	Stored propellant
Nuclear fission rocket		TFD		Stored H ₂
Solar-heated rocket			TFD	Stored H ₂
Photon rocket (big light bulb)		TFND		Photon ejection (no stored propellant)
Solar sail			TFD	Photon reflection (no stored propellant)

^aD/P developed and/or considered practical; TFD, technical feasibility has been demonstrated, but development is incomplete; TFND, technical feasibility has not yet been demonstrated.

input modes in rocket propulsion systems are either heat or electricity. Useful output thrust comes from the kinetic energy of the ejected matter and from the propellant pressure on inner chamber walls and at the nozzle exit; thus, rocket propulsion systems primarily convert input energies into the kinetic energy of the exhausted gas. The ejected mass can be in a solid, liquid, or gaseous state. Often, combinations of two or more phases are ejected. At very high temperatures, ejected matter can also be in a plasma state, which is an electrically conducting gas.

1.1. DUCT JET PROPULSION

This class, commonly called *air-breathing engines*, comprises devices which entrain and energize air flow inside a duct. They use atmospheric oxygen to burn fuel stored in the flight vehicle. This class includes turbojets, turbofans, ramjets, and pulse-jets. These are mentioned here primarily to provide a basis for comparison with rocket propulsion and as background for combined rocket–duct engines, which are mentioned later. Table 1–2 compares several performance characteristics of specific chemical rockets with those of typical turbojets and ramjets. A high specific impulse (which is a measure of performance to be defined later) relates directly to long-flight ranges and thus indicates the superior range capability of air-breathing engines over chemical rocket propulsion systems at relatively low earth altitudes. However, the uniqueness of rocket propulsion systems (for example, high thrust to weight, high thrust to frontal area, and thrust nearly independent of altitude) enables flight in rarefied air and exclusively in space environments.

TABLE 1–2. Comparison of Several Characteristics of a Typical Chemical Rocket Propulsion System and Two-Duct Propulsion Systems

Feature	Chemical Rocket Engine or Rocket Motor	Turbojet Engine	Ramjet Engine
Thrust-to-weight ratio, typical	75:1	5:1, turbojet and afterburner	7:1 at Mach 3 at 30,000 ft
Specific fuel consumption (pounds of propellant or fuel per hour per pound of thrust) ^a	8–14	0.5–1.5	2.3–3.5
Specific thrust (pounds of thrust per square foot frontal area) ^b	5000–25,000	2500 (low Mach ^c numbers at sea level)	2700 (Mach 2 at sea level)
Specific impulse, typical ^d (thrust force per unit propellant or fuel weight flow per second)	270 sec	1600 sec	1400 sec
Thrust change with altitude	Slight increase	Decreases	Decreases
Thrust vs. flight speed	Nearly constant	Increases with speed	Increases with speed
Thrust vs. air temperature	Constant	Decreases with temperature	Decreases with temperature
Flight speed vs. exhaust velocity	Unrelated, flight speed can be greater	Flight speed always less than exhaust velocity	Flight speed always less than exhaust velocity
Altitude limitation	None; suited to space travel	14,000–17,000 m	20,000 m at Mach 3 30,000 m at Mach 5 45,000 m at Mach 12

^aMultiply by 0.102 to convert to kg/(hr-N).^bMultiply by 47.9 to convert to N/m².^cMach number is the ratio of gas speed to the local speed of sound (see Eq. 3–22).^d*Specific impulse* is a performance parameter defined in Chapter 2.

The *turbojet engine* is the most common of ducted engines. Figure 1–1 shows its basic elements.

For supersonic flight speeds above Mach 2, the *ramjet engine* (a pure duct engine) becomes possible for flights within the atmosphere. Compression is purely gas dynamic and thrust is produced by increasing the momentum of the subsonic compressed air as it passes through the ramjet, basically as is accomplished in the turbojet and turbofan engines but without any compressor or turbine hardware. Figure 1–2 shows the basic components of a ramjet. Ramjets with subsonic combustion and hydrocarbon fuels have an upper speed limit of approximately Mach 5; hydrogen fuel, with hydrogen cooling, raises this to at least Mach 16. Ramjets with supersonic combustion, known as *scramjets*, have flown in experimental vehicles. All ramjets

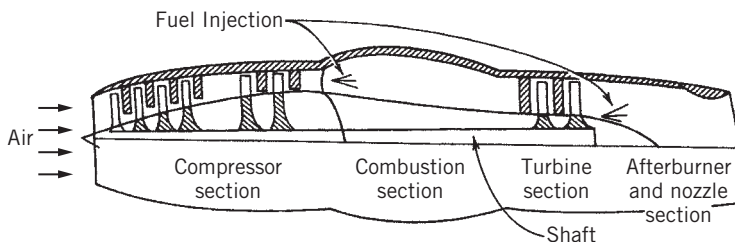


FIGURE 1-1. Simplified schematic diagram of a turbojet engine.

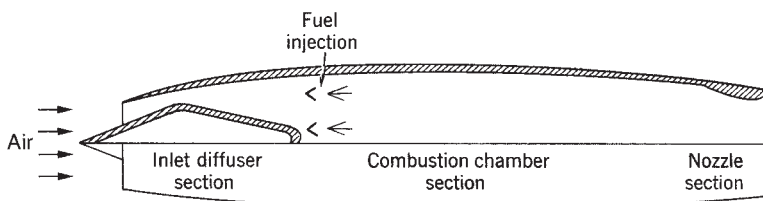


FIGURE 1-2. Simplified diagram of a ramjet with a supersonic inlet (a converging/diverging flow passage).

must depend on rocket or aircraft boosters for initial acceleration to supersonic conditions and operating altitudes, and on oblique shocks to compress and decelerate the entrance air. Applications of ramjets with subsonic combustion include shipboard- and ground-launched anti-aircraft missiles. Studies of a hydrogen-fueled ramjet for hypersonic aircraft looked promising, but as of this writing they have not been properly demonstrated; one supersonic flight vehicle concept combines a ramjet-driven high-speed airplane and a one- or two-stage rocket booster for driving the vehicle to its operating altitude and speed; it can travel at speeds up to a Mach number of 25 at altitudes of up to 50,000 m.

No truly new or significant rocket technology concepts have been implemented in recent years, reflecting a certain maturity in this field. Only a few new applications for proven concepts have been found, and those that have reached production are included in this edition. The culmination of research and development efforts in rocket propulsion often involves adaptations of new approaches, designs, materials, as well as novel fabrication processes, cost, and/or schedule reductions to new applications.

1.2. ROCKET PROPULSION

Rocket propulsion systems may be classified in a number of ways, for example, according to energy source type (chemical, nuclear, or solar) or by their basic function (booster stage, sustainer or upper stages, attitude control, orbit station keeping, etc.)

or by the type of vehicle they propel (aircraft, missile, assisted takeoff, space vehicle, etc.) or by their size, type of propellant, type of construction, and/or by the number of rocket propulsion units used in a given vehicle.

Another useful way to classify rockets is by the method of producing thrust. The thermodynamic expansion of a gas in a supersonic nozzle is utilized in most common rocket propulsion concepts. The internal energy of the propellant is converted into exhaust kinetic energy, and thrust is also produced by the pressure on surfaces exposed to the exhaust gases, as will be shown later. This same thermodynamic theory and the same generic equipment (i.e., a chamber plus a nozzle) is used for jet propulsion, rocket propulsion, nuclear propulsion, laser-thermal and solar-thermal propulsion, and in some types of electrical propulsion. Totally different methods of producing thrust are used in nonthermal types of electric propulsion. As described below, these electric systems use magnetic and/or electric fields to accelerate electrically charged atoms or molecules at very low gas densities. It is also possible to obtain very small accelerations by taking advantage of the difference in gravitational attraction as a function of earth altitude, but this method is not treated in this book.

The Chinese developed and used solid propellant in rocket missiles over 800 years ago, and military “bombardment rockets” were used frequently in the eighteenth and nineteenth centuries. However, the most significant developments of rocket propulsion took place in the twentieth century. Early pioneers included the Russian Konstantin E. Ziolkowsky, who is credited with the fundamental rocket flight equation and his 1903 proposals to build rocket vehicles. Robert H. Goddard, an American, is credited with the first flight using a liquid propellant rocket engine in 1926. For the history of rockets, see Refs. 1–1 to 1–7.

Chemical Rocket Propulsion

Energy from the combustion reaction of chemical propellants, usually a fuel and an oxidizer, in a high-pressure chamber goes into heating reaction product gases to high temperatures (typically 2500 to 4100 °C or 4500 to 7400 °F). These gases are subsequently expanded in a supersonic nozzle and accelerated to high velocities (1800 to 4300 m/sec or 5900 to 14,100 ft/sec). Since such gas temperatures are about twice the melting point of steel, it is necessary to cool or insulate all the surfaces and structures that are exposed to the hot gases. According to the physical state of the stored propellant, there are several different classes of chemical rocket propulsion devices.

Liquid propellant rocket engines use propellants stored as liquids that are fed under pressure from tanks into a *thrust chamber*.^{*} A typical pressure-fed liquid propellant rocket engine system is schematically shown in Fig. 1–3. The *bipropellant* consists of a liquid oxidizer (e.g., liquid oxygen) and a liquid fuel (e.g., kerosene). A *monopropellant* is a single liquid that decomposes into hot gases when properly catalyzed.

^{*}The term *thrust chamber*, used for the assembly of the injector, nozzle, and chamber, is preferred by several official agencies and therefore has been used in this book. For small spacecraft control rockets the term *thruster* (a small thrust chamber) is commonly used, and this term will be used in some sections of this book.

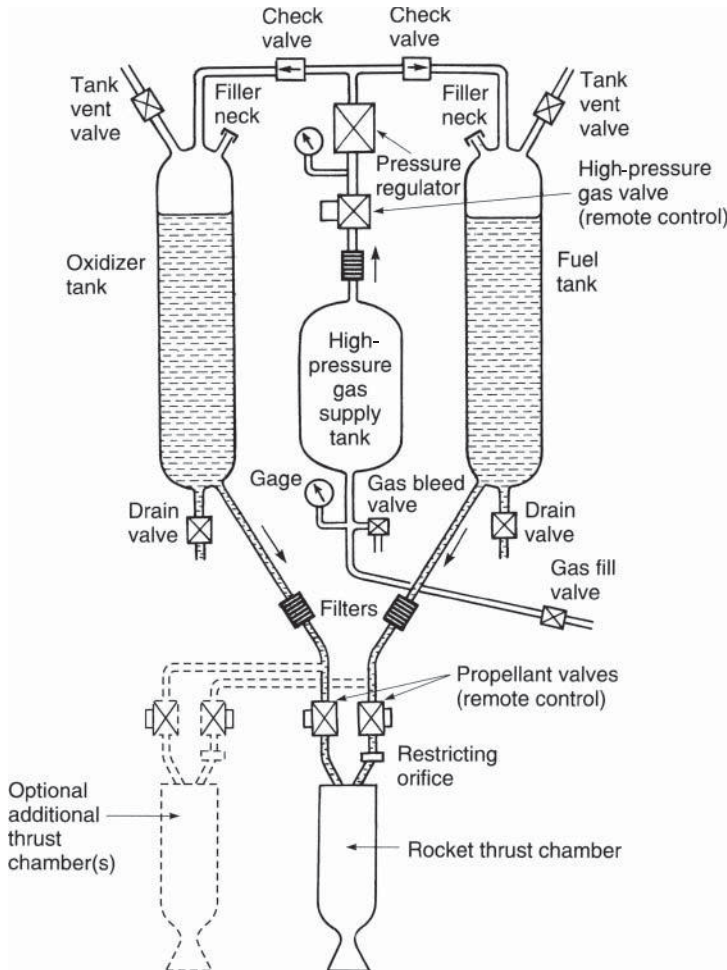


FIGURE 1–3. Schematic flow diagram of a liquid propellant rocket engine with a gas pressure feed system. The dashed lines show a second thrust chamber, but some engines have more than a dozen thrust chambers supplied by the same feed system. Also shown are components needed for start and stop, controlling tank pressure, filling propellants and pressurizing gas, draining or flushing out remaining propellants, tank pressure relief or venting, and several sensors.

Gas pressure feed systems are used mostly on low-thrust, low-total-energy propulsion systems, such as those used for attitude control of flying vehicles, often with more than one thrust chamber per engine. The larger bipropellant rocket engines use one or more turbopump-fed liquids as shown in Fig. 1–4. Pump-fed liquid rocket systems are most common in applications needing larger amounts of propellant and higher thrust, such as those in space launch vehicles. See Refs. 1–1 to 1–6.

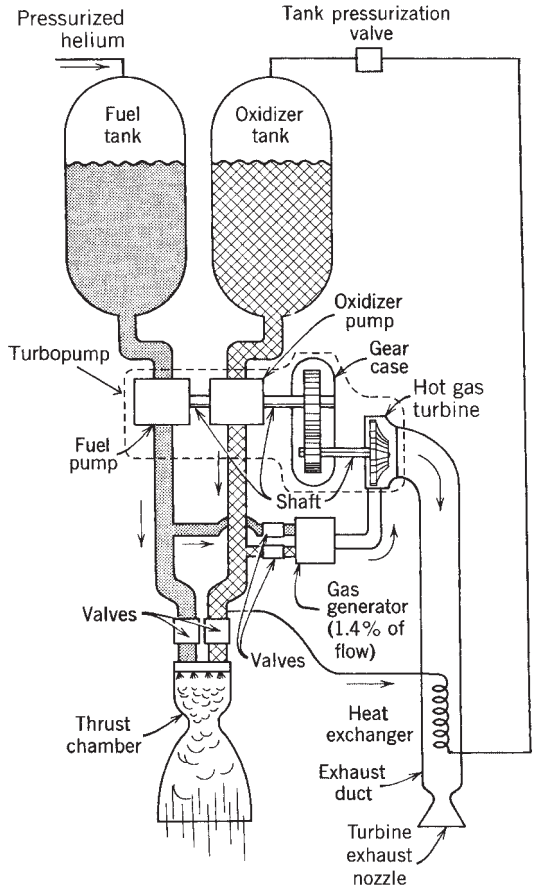


FIGURE 1-4. Simplified schematic diagram of a liquid propellant rocket engine with one type of turbopump feed system and a separate gas generator, which generates “warm” gas for driving the turbine. Not shown are components necessary for controlling the operation, filling, venting, draining, or flushing out propellants, filters, pilot valves, or sensors. This turbopump assembly consists of two propellant pumps, a gear case, and a high-speed turbine. Many turbopumps have no gear case.

Chemical propellants react to form hot gases inside the *thrust chamber* and proceed in turn to be accelerated through a supersonic nozzle from which they are ejected at a high velocity, thereby imparting momentum to the vehicle. Supersonic nozzles consist of a converging section, a constriction or throat, and a conical or bell-shaped diverging section, as further described in the next two chapters.

Some liquid rocket engines permit repetitive operation and can be started and shut off at will. If the thrust chamber is provided with adequate cooling capacity, it is possible to run liquid rockets for hours, depending only on the propellant supply. A liquid rocket propulsion system requires several precision valves and some have

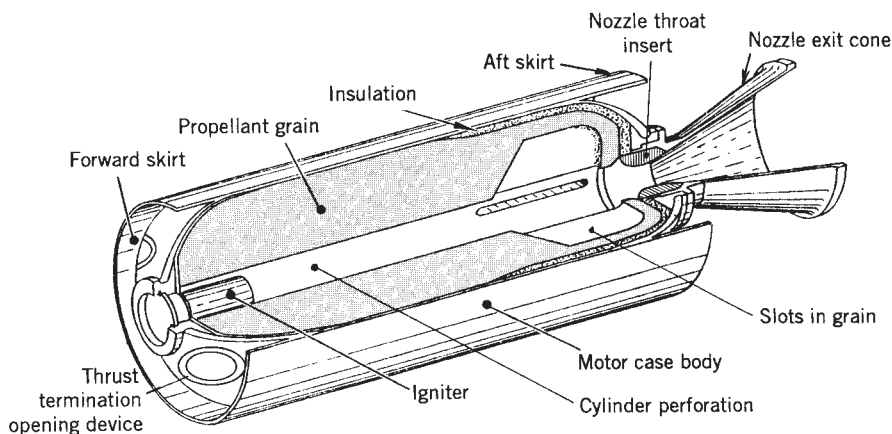


FIGURE 1–5. Simplified perspective three-quarter section view of a typical solid propellant rocket motor with the propellant grain bonded to the case and to the insulation layer, and with a conical exhaust nozzle. The cylindrical case with its forward and aft hemispherical domes forms a pressure vessel containing the combustion chamber pressure. Adapted with permission from Ref. 12–1.

complex feed mechanisms that include propellant pumps, turbines and gas generators. All have propellant-pressurizing devices and relatively intricate combustion/thrust chambers.

In *solid propellant rocket motors** the ingredients to be burned are already stored within a combustion chamber or *case* (see Fig. 1–5). The solid propellant (or charge) is called the *grain*, and it contains all the chemical elements for complete burning. Once ignited, it is designed to burn smoothly at a predetermined rate on all the exposed internal grain surfaces. In the figure, initial burning takes place at the internal surfaces of the cylinder perforation and at the four slots. The internal cavity expands as propellant is burned and consumed. The resulting hot gases flow through the supersonic nozzle to impart thrust. Once ignited, motor combustion is designed to proceed in an orderly manner until essentially all the propellant has been consumed. There are no feed systems or valves. See Refs. 1–6 to 1–9.

Liquid and solid propellants, together with the propulsion systems that use them, are discussed in Chapters 6 to 11 and 12 to 15, respectively. Rocket propulsion selection systems, for both liquid and solid propellants, are compared in Chapter 19.

Gaseous propellant rocket engines use a stored high-pressure gas, such as air, nitrogen, or helium, as working fluid. Such stored gases require relatively heavy tanks. These cold gas thrusters were used in many early space vehicles for low-thrust maneuvers and for attitude-control systems, and some are still used today. Heating the gas with electrical energy or by the combustion of certain monopropellants in a chamber

*Historically, the word *engine* is used for a liquid propellant rocket propulsion system and the word *motor* is used for solid propellant rocket propulsion. They were developed originally by different groups.

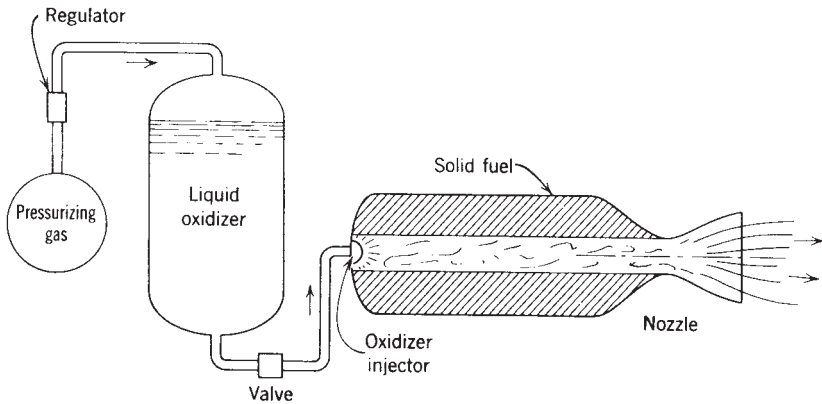


FIGURE 1–6. Schematic diagram of a typical hybrid rocket engine. The relative positions of the oxidizer tank, high-pressure gas tank, and the fuel chamber with its nozzle depend on the particular vehicle design.

improves their performance, and this has often been called *warm gas propellant rocket propulsion*. Chapter 7 reviews gaseous propellants.

Hybrid propellant rocket propulsion systems use both liquid and solid propellant storage. For example, if a liquid oxidizing agent is injected into a combustion chamber filled with a solid carbonaceous fuel grain, the chemical reaction produces hot combustion gases (see Fig. 1–6). They are described further in Chapter 16. Several have flown successfully.

Combinations of Ducted Jet Engines and Rocket Engines

The Tomahawk surface-to-surface missile uses a sequenced two-stage *propulsion system*. The solid propellant rocket booster lifts the missile away from its launch platform and is discarded after its operation. A small turbojet engine then sustains their low-level flight at nearly constant subsonic speed toward the target.

Ducted rocket propulsion systems, sometimes called *air-augmented rocket propulsion systems*, combine the principles of rocket and ramjet engines; they give higher performance (specific impulse) than chemical rocket engines but can only operate within the earth’s atmosphere. Usually, the term *air-augmented rocket* denotes mixing of air with the rocket exhaust (made fuel rich for afterburning) in proportions that enable the propulsion device to retain those characteristics that typify rocket engines, for example, high static thrust and high thrust-to-weight ratio. In contrast, the ducted rocket is often like a ramjet in that it must be boosted to operating speed and uses the rocket components more as a fuel-rich gas generator (liquid or solid).

The action of rocket propulsion systems and ramjets can be combined. An example of these two are propulsion systems operating in sequence and then in tandem and yet utilizing a common combustion chamber volume, as shown in Fig. 1–7. Such a low-volume configuration, known as an *integral rocket–ramjet*, has been attractive

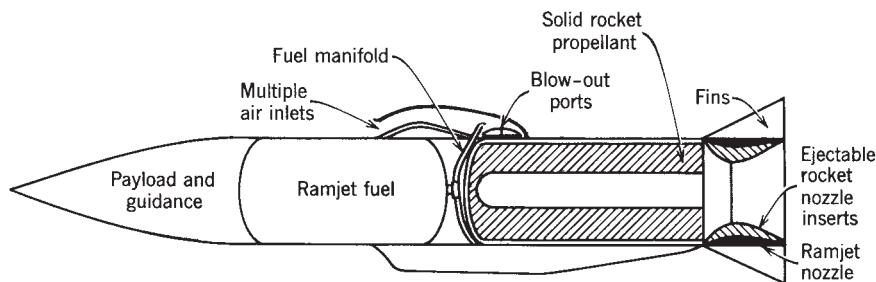


FIGURE 1–7. Simplified diagram of an air-launched missile with integral rocket–ramjet propulsion. After the solid propellant has been consumed in boosting the vehicle to flight speed, the rocket combustion chamber becomes the ramjet combustion chamber with air burning the ramjet liquid fuel. Igniter and steering mechanisms are not shown.

for air-launched missiles using ramjet propulsion. The transition from rocket engine to ramjet requires enlarging the exhaust nozzle throat (usually by ejecting rocket nozzle parts), opening the ramjet air inlet–combustion chamber interface, and following these two events with a normal ramjet starting sequence.

A *solid fuel ramjet* uses grains of solid fuel that gasify or ablate and then react with air. Good combustion efficiencies have been achieved with a patented boron-containing solid fuel fabricated into grains similar to a solid propellant rocket motor and burning in a manner similar to a hybrid rocket propulsion system.

Nuclear Rocket Engines

These are basically a type of liquid propellant rocket engine where the power input comes from a single nuclear reactor and not from any chemical combustion. During the 1960s an experimental rocket engine with a nuclear fission graphite reactor was built and ground tested with liquid hydrogen as the propellant. It delivered an equivalent altitude specific impulse (this performance parameter is explained in Chapter 2) of 848 sec, a thrust of over 40,000 lbf at a nuclear reactor power level of 4100 MW with a hydrogen temperature of 2500 K. No further ground tests of nuclear fission rocket engines have been undertaken.

Public concerns about any ground and/or flight accident with the inadvertent spreading of radioactive materials in the Earth’s environment have caused the termination of nuclear rocket engine work. It is unlikely that nuclear rocket engines will be developed in the next few decades and therefore no further discussion is given in here. Our Eighth Edition has additional information and references on nuclear propulsion.

Electric Rocket Propulsion

Electric propulsion has been attractive because of its comparatively high performance, producing desired amounts thrust with moderately low propellant utilization,

but they are limited to relatively low thrusts by existing electrical power supplies. This type of propulsion is much too low for earth launches and atmospheric flight because it requires rather massive and relatively inefficient power sources (but in spacecraft they can often be shared with other subsystems). Unlike chemical propulsion, electric propulsion utilizes energy sources (nuclear, solar radiation, or batteries) not contained in the propellant being utilized. The thrust is usually quite low, levels typical of orbit maintenance (0.005 to 1 N). In order to accomplish significant increases in vehicle velocity, it is necessary to apply such low thrusts (and their small accelerations) during times considerably longer than with chemical propulsion, some for months and even years (see Ref. 1–10 and Chapter 17).

Of the three basic electric types, *electrothermal* rocket propulsion most resembles the previously discussed liquid-propellant chemical rocket units; a propellant is heated electrically (with solid resistors called resistojets or electric arcs called arcjets), and the hot gas is then thermodynamically expanded through a supersonic nozzle (see Fig. 1–8). These electrothermal thrusters typically have thrust ranges of 0.01 to 0.5 N, with exhaust velocities of 1000 to 7800 m/sec; they use ammonium, hydrogen, nitrogen, or hydrazinede composition products as the propellant.

The two other types—the *electrostatic* or *ion propulsion* thrusters and the *electromagnetic* or *magnetoplasma* thrusters—accomplish propulsion by different means, and no thermodynamic gas expansion in a nozzle is necessary. Both work only in a vacuum. In an ion thruster (see Fig. 1–9) the working fluid (typically, xenon) is ionized by stripping off electrons, and then the heavy ions are accelerated to very high velocities (2000 to 60,000 m/sec) by means of electrostatic fields. The ions are subsequently electrically neutralized by combining them with emitted electrons to prevent the buildup of a “space charge” on the vehicle.

In electromagnetic thrusters, a plasma (an energized gaseous mixture of ions, electrons, and neutral particles) is accelerated by the interaction between electric currents and perpendicular magnetic fields and this plasma is then ejected at high velocities (1000 to 75,000 m/sec). There are many different types and geometries.

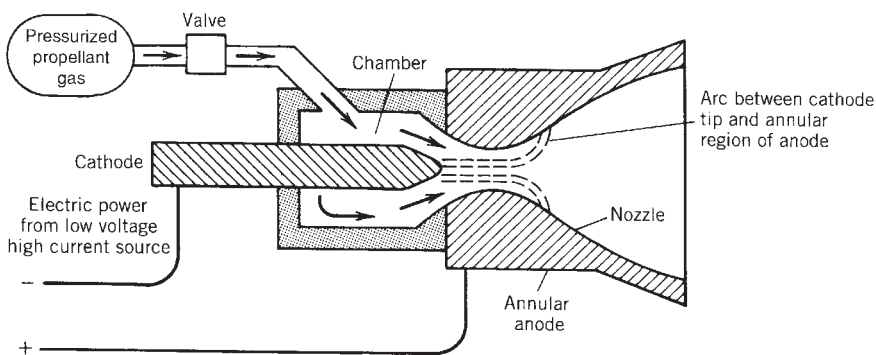


FIGURE 1–8. Simplified schematic diagram of arcjet thruster. The arc plasma temperature is very high (perhaps 15,000 K) and the anode, cathode, and chamber will get hot (1000 K) due to heat transfer.

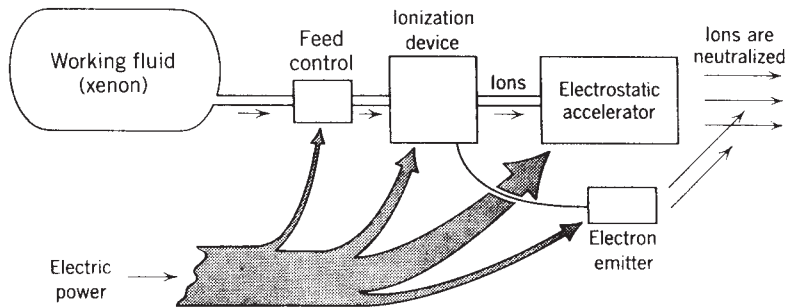


FIGURE 1–9. Simplified schematic diagram of a typical ion thruster, showing the approximate distribution of the electric power by the width of the arrow.

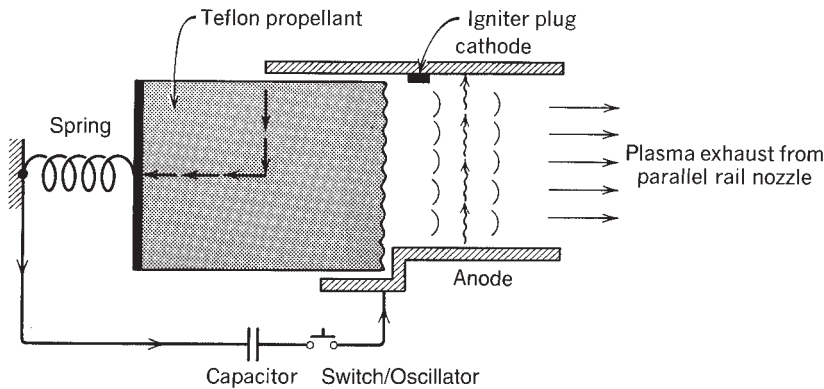


FIGURE 1–10. Simplified diagram of a pulsed plasma thruster with self-induced magnetic acceleration. When the capacitor is discharged, an arc is struck at the left side of the rails. The high arc current closes the loop, thus inducing a magnetic field. The action of the current and the magnetic field causes the plasma to be accelerated at right angles to both the magnetic field and the current in the direction of the rails. Each time the arc is created, a small amount of solid propellant (Teflon) is vaporized and converted to a small plasma cloud, which (when ejected) gives a small pulse of thrust. Actual units can operate with many pulses per second.

The Hall-effect thruster, a relatively new entry, may also be considered as electrostatic (see Chapter 17). A simply configured, pulsed electrical thruster with a solid (stored) propellant is shown in Fig. 1–10; it has been used for spacecraft attitude control (Ref. 1–10).

Other Rocket Propulsion Concepts

One concept is the *solar thermal rocket*; it has large-diameter optics to concentrate the sun’s radiation (e.g., with lightweight precise parabolic mirrors or Fresnel lenses) onto a receiver or optical cavity, see Ref. 1–11. Figure 1–11 shows one embodiment

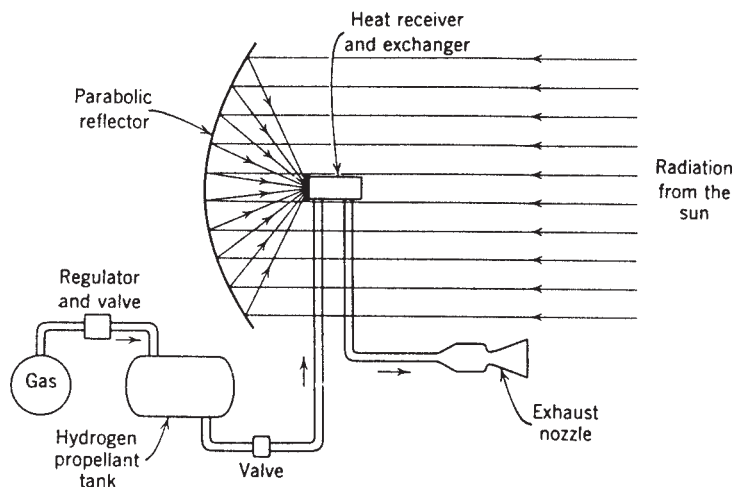


FIGURE 1–11. Schematic diagram of a solar thermal rocket concept.

and some data is given in Table 2–1. The receiver is made of high-temperature metal (such as tungsten or rhenium) and has a heat exchanger that heats the working fluid, usually hydrogen, up to perhaps 2500 °C; the hot gas is then exhausted through one or more nozzles. The large reflective mirror must be pointed toward the sun, and this requires orientation adjustments if the spacecraft orbits around the Earth or other planets. Performance can be two to three times higher than that of chemical rockets and thrust levels are low (1 to 10 N). Since large lightweight optical elements cannot withstand drag forces without deformation, such units are deployed outside the atmosphere. Contamination is negligible, but storage or refueling of liquid hydrogen is a challenge. Problems being investigated include rigid, lightweight mirror or lens structures; operational lifetimes; and minimizing hydrogen evaporation and heat losses to other spacecraft components. An experimental solar-thermal rocket propulsion system flew in a satellite in 2012. It has not been approved for a production application (as of 2015) to the authors' knowledge.

The *solar sail* is another concept. It is basically a large photon reflecting surface. The power source for the solar sail is external to the vehicle (see Ref. 1–12), but the vehicle can only move away from the sun. Concepts for transmitting radiation energy (by lasers or microwaves) from ground stations on Earth to satellites have been proposed but not yet tested.

International Rocket Propulsion Effort

Active development and/or production of rocket propulsion systems have been under way in more than 30 different countries. A few foreign rocket units are mentioned in this book together with their characteristics and with references to the international rocket literature. Although most of the data in this book are taken from U.S. rocket experiences, this is not intended to minimize the significance of foreign achievements.

At the time of this writing, the only joint major international program has been the *International Space Station* (ISS), a multiyear cooperative effort with major contributions from the United States and Russia and active participation by several other nations. This manned orbital space station is used for conducting experiments and observations on a number of research projects. See Ref. 1–13.

1.3. APPLICATIONS OF ROCKET PROPULSION

Because rocket propulsion can reach performances unequaled by other prime movers, it has its own field of applications and does not usually compete with other propulsion devices. Selection of the best rocket propulsion system type and design for any given application is a complex process involving many factors, including system performance, reliability, cost, propulsion system size, and compatibility, as described in Chapter 19. Examples of important applications are given below and some are discussed further in Chapter 4.

Space Launch Vehicles

Since 1957 there have been numerous space launch attempts with a better than 95% success record. *Space launch vehicles* or *space boosters* can be broadly classified as expendable or recoverable/reusable, by the type of propellant (storable or cryogenic liquid or solid propellants), number of stages (single-stage, two-stage, etc.), size/mass of payloads or vehicles, and as manned or unmanned. There are many different *missions* and *payloads* for space launch vehicles. Discussed below are the following categories: commercial missions (e.g., communications), military missions (e.g., reconnaissance), nonmilitary missions (e.g., weather observation), and space exploration missions (e.g., flights to the planets).

Each space launch has a specific space flight objective, such as an Earth orbit or a moon landing. See Ref. 1–14. It uses between two and five stages, each with its own propulsion systems and each usually fired sequentially after the lower stage is expended. Selection for the number of stages is based on the specific space trajectory, the number and types of maneuvers, the energy content of a unit mass of the propellant, the payload size, as well as other factors. The initial stage, usually called the booster stage, is the largest; this stage is then separated from the ascending vehicle before the second-stage propulsion system is ignited and operated. As explained in Chapter 4, adding extra stages may permit significant increases in the payload (such as more scientific instruments or more communications gear).

Each stage of a multistage launch vehicle is essentially a complete vehicle in itself and carries its own propellant, its own rocket propulsion system or systems, and its own control system. Once the propellant of a given stage is expended, its remaining mass (including empty tanks, cases, structure, instruments, etc.) is no longer useful to succeeding stages. By dropping off this mass, it is possible to accelerate the final stage with its payload to a higher terminal velocity than would be attained if multiple

staging were not used. Both solid propellant and liquid propellant rocket propulsion systems have been amply utilized in low Earth orbit missions.

Figure 1–12 shows the Delta IV HEAVY lift space launch vehicle at takeoff. Its propellants are liquid oxygen/liquid hydrogen (LOX/LH₂) in all its main engines. Its booster engine, the Aerojet Rocketdyne RS-68A, is shown in Figs. 6–9a and 6–9b and data is in Table 11–2; its second-stage engine, the Aerojet Rocketdyne RL-10B-2 LOX/LH₂ (24,750 lbf thrust) is shown in Fig. 8–17 and data is in Table 8–1. The two liquid propellant strap-on booster pods (with the same booster engine) are removed for launching smaller payloads. Figure 1–13 shows the Atlas V space launch vehicle. Its booster engine is the Russian (Energomash) RD-180, it has Aerojet solid propellant strap-on boosters, and the upper stage engine is the Aerojet Rocketdyne RL 10A-4-2 LOX/LH₂ engine. The Russian (Energomash) LOX/kerosene RD-180 engine is shown in Ref. 1–2 as its Figure 7.10–11 and data is in its Table 7.10–2. In both of these launch vehicles the payload is carried on top of the second stage, which has its own propulsion set of small thrusters. Table 1–3 gives data for the larger propulsion systems in these two U.S. launch vehicles. Not shown in Table 1–3 are two additional stage separation systems for the Delta IV HEAVY space launch vehicle. One consists of a set of small solid propellant rocket motors that are installed in the two outboard boosters; one quarter of these motors are installed under each of the nozzle housings and one quarter under each nose fairing at the outboard booster stages. Their purpose is to move two outboard boosters (just after thrust termination) from the center or core booster (which continues to operate) and thus prevent any collision. These separation solid propellant motor boosters have a relatively high thrust of very short duration.

Also not listed in Table 1–3 is an attitude control system for the second stage of the Delta IV HEAVY. It has 12 small restartable monopropellant hydrazine thrusters that provide pitch, yaw, and roll control forces to this upper stage during the powered flight (from the RL-10B-2) and during the unpowered portion of this upper stage. These thrusters and the separation motors are not evident in Fig. 1–12.

The U.S. *Space Shuttle*, which was retired in 2011, provided the first *reusable spacecraft* that could glide and land on a runway. Figure 1–14 shows the basic configuration of the Space Shuttle at launch, which consisted of two stages, the booster, the orbiter stage, and an external tank. It shows all the 67 rocket propulsion systems of the shuttle. These consisted of 3 main engines (LOX/LH₂ of 470,000 lbf vacuum thrust each, see Chapter 7 for chemical nomenclature), 2 orbital maneuvering engines (N₂O₄/MMH of 6,000 vacuum thrust each), 38 reaction control primary thrusters (N₂O₄/MMH of 870 lbf vacuum thrust each), 6 reaction control Vernier thrusters (N₂O₄/MMH of 25 lbf vacuum thrust each), 2 large segmented booster solid propellant motors (composite solid propellant of 3.3×10^6 lbf thrust at sea level each), and 16 stage separation rocket motors (with composite solid propellants of 22,000 lbf vacuum thrust each, operating for 0.65 sec); they were activated after thrust termination of the boosters in order to separate them from the external tank. The orbiter was the reusable vehicle, a combination space launch vehicle, spacecraft, and glider for landing. The two solid propellant strap-on rocket motors were then



FIGURE 1–12. Delta IV HEAVY lift space launch vehicle. The center liquid propellant booster stage has a Pratt & Whitney Rocketdyne RS-68A rocket engine (LOX/LH₂). The two strap-on stages each use the same engine. See Figs. 6–9a and 6–9b. Courtesy Aerojet-Rocketdyne.



FIGURE 1–13. Atlas V space launch vehicle with three (or five) strap-on stages using Aerojet solid rocket motors and a central Energomash (Russia) RD-180 liquid propellant booster rocket engine running on LOX/kerosene. See Table 1–3 for key parameters. Courtesy United Launch Alliance.

TABLE 1–3. Major Propulsion Systems for Two U.S. Launch Vehicles

Vehicle	Propulsion System Designation (Propellant)	Stage	No. of Propulsion Systems per Stage	Thrust (lbf/kN) per Engine/Motor	Specific Impulse (sec)	Mixture		Chamber Pressure (psia)	Nozzle Exit Area Ratio	Inert Engine Mass (lbm/kg)
						Ratio, Oxidizer to Fuel Flow				
DELTA IV HEAVY	RS-68A (LOX/LH ₂)	1	1 or 3	797,000/3548 ^a	411 ^a	5.97		1557	21.5:1	14,770/6,699
	RL 10B-2 (LOX/LH ₂)	2	1	702,000/3123 ^{b, c}	362 ^b	5.88		633	285:01:00	664 lbm
ATLAS V	Solid Booster	0	Between 1 ^d and 5	24,750/0.110 ^a 287,346/1.878 ^b each	465.5 ^a 279.3 ^b	N/A		3722	16:1 ^c	102,800 lbm (loaded)
	RD-180 ^d (LOX/Kerosene)	1	1	933,400/4.151 ^a	310.7 ^b	2.72		610	36.4:1	12,081/5,480
	RD 10A-4-2 (LOX/LH ₂)	2	1 or 2	860,200/3.820 ^{b, c}	337.6 ^a	4.9–5.8			84.1:1	5330 kg
				22,300/99.19 ^a	450.5 ^a					370/168

^aVacuum value.
^bSea-level value.
^cAt ignition.
^dRussian RD-180 engine has 2 gimbal-mounted thrust chambers.

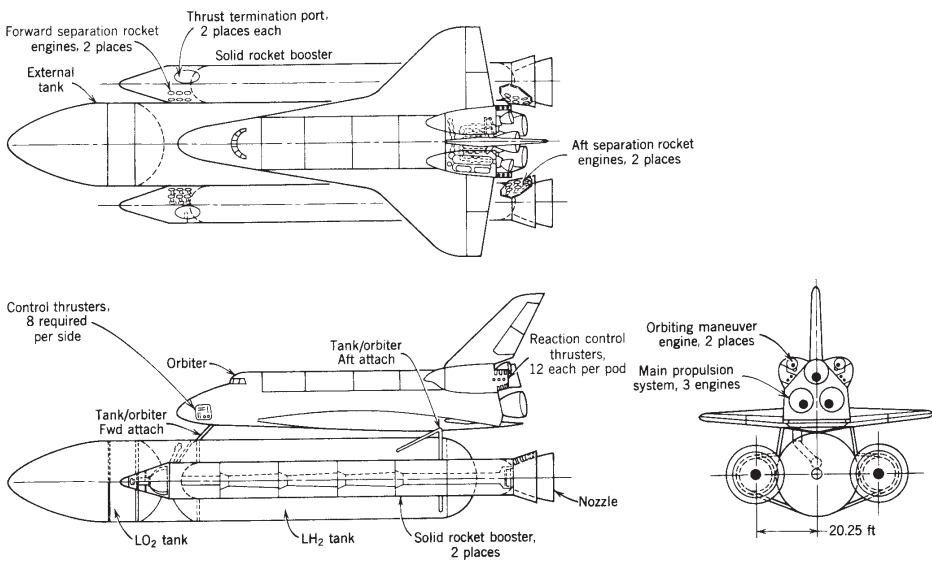


FIGURE 1–14. Simplified diagrams of the original Space Shuttle vehicle. The shuttle orbiter—the delta-winged vehicle about the size of a medium-range jet liner—was a reusable, cargo-carrying, spacecraft–airplane combination that took off vertically and landed horizontally as a glider. Each shuttle orbiter was designed for a minimum of 100 missions and could carry as much as 65,000 lb of payload to a low Earth orbit, and a crew of up to four members. It could return up to 25,000 lb of payload back to earth. NASA’s new manned Space Launch System plans using a modified Space Shuttle engine (see Fig. 6–1) designated as the RS-25.

the largest in existence; they were equipped with parachutes for sea recovery of the burned-out motors. The large LO_2/LH_2 external tank was jettisoned and expended just before orbit insertion (see Ref. 1–15). The Space Shuttle accomplished both civilian and military missions, placing astronauts and satellites in orbit, undertaking scientific exploration, supplying the International Space Station, and repairing, servicing, and retrieving satellites. Today, the retired orbiters (the winged vehicles of the Space Shuttle) are on museum displays. At the time of this writing (2015), the National Aeronautics and Space Administration (NASA) had awarded the initial research and development (R&D) contracts in several critical areas for a new large manned space flight vehicle identified as Orion and as SLS (Space Launch System).

A new generation of manned space flight vehicles is currently being developed by several entrepreneurial U.S. and foreign companies, some with private capital. They are aimed at future commercial markets that include sending tourists into space forays. All are based on reusable spacecraft, some with reusable launch vehicles, some with vertical and some with horizontal takeoff or landing. A number of suborbital flights have already been accomplished by test pilots using new winged vehicles. It is too early to predict which of these organizations will be successful in commercializing manned space flight. Some new companies have already developed structures and engines for practical “low-cost” launch vehicles; these are, at the time of this writing, being routinely used for transporting supplies to the International Space Station (which is in a low Earth orbit). To bring down costs, the first stage of future launch vehicles will need to be recoverable and reusable.

A single-stage-to-orbit vehicle, an attractive concept because it avoids the costs and complexities of staging, would be expected to have improved reliability (simpler structures, fewer components). However, its envisioned payload has been too small for economic operation. To date efforts at developing a rocket-propelled single-stage-to-orbit vehicle have not been successful.

Spacecraft

Depending on their mission, *spacecraft* can be categorized as Earth satellites, lunar, interplanetary, and trans-solar types, and/or as manned and unmanned spacecraft. Reference 1–16 lists over 20,000 satellites and categorizes them as satellites for communications, weather, navigation, scientific exploration, deep space probes, observation (including radar surveillance), reconnaissance, and other applications. Rocket propulsion is needed for both *primary propulsion* (i.e., acceleration along the flight path, such as for ascents, orbit insertion, or orbit change maneuvers) and for secondary propulsion functions in these vehicles. Some of the *secondary propulsion* functions are attitude control, spin control, momentum wheel and gyro unloading, rendezvous in space, stage separation, and for the settling of liquids in tanks. Spacecraft need a variety of different rocket propulsion systems and some thrusters can be very small. For spacecraft attitude control about three perpendicular axes, each in two rotational directions, the system must allow the application of pure torque for six modes of angular freedom, thus requiring a minimum of 12 thrusters. Some missions require as few as 4 to 6 thrusters, whereas the more complex manned

spacecraft have 20 to 50 thrusters in all of their stages. Often, the small *attitude control rockets* must give pulses or short bursts of thrust, necessitating thousands of restarts. See Section 6.7 and Ref. 1–17.

A majority of spacecraft and space launch vehicles use liquid propellant engines for principal propulsion because of their better performance. Liquid propellants are used as both primary and secondary propulsion systems. A few vehicles have used solid propellant rocket motors for booster stages and some for orbit injection. Some spacecraft operate successfully with electrical propulsion for attitude control. Recently electrical propulsion (EP) systems have also been used for some primary and secondary spacecraft propulsion missions in long-duration space flights. Because of their low thrust, space operations/maneuvers with EP require relatively longer times to reach desired velocity increases. For example, transfer from a low Earth orbit to a geosynchronous orbit may take as long as two or three months compared to a few hours with chemical propulsion systems. See Chapter 17. Designs utilizing all-electric propulsion systems for upper stages are a relatively recent trend.

Micropropulsion is a new designation for thrust levels applicable to small spacecraft of less than 100 kg, or 220 lbm. See Ref. 1–18. It encompasses a variety of different propulsion concepts, such as certain very low thrust liquid mono- and bipropellant rocket engines, small gaseous propellant rocket engines, several types of electrical propulsion systems, and emerging advanced versions of these. Many are based on recent fabrication techniques for very small components (valves, thrusters, switches, insulators, or sensors) by micromachining and electromechanical processes.

Military and Other Applications

Military applications can be classified as shown in Table 1–4. Rocket propulsion for new U.S. missiles is presently based almost exclusively on solid propellant rocket motors. These can be *strategic missiles*, such as long-range ballistic missiles (800 to 9000 km range), which are aimed at military targets within an enemy country, or *tactical missiles*, which are intended to support or defend military ground forces, aircraft, or navy ships. Table 1–5 shows some preferred rocket propulsion systems for selected applications.

The term *surface launch* can mean launching from the ground, the ocean surface (from a ship), or from underneath the sea (submarine launch). Some tactical missiles, such as the air-to-surface short-range attack missile (SRAM), have a two-pulse solid propellant motor, where two separate insulated grains of different solid properties are in the same motor case; the time interval before starting the second pulse can be timed to control the flight path or speed profile. Many countries now have tactical missiles in their military inventories, and many of these countries have the capability to produce their own vehicles and their rocket propulsion systems.

Applications that were popular 40 to 70 years ago but are no longer active include liquid propellant rocket engines for propelling military fighter aircraft to altitude, assisted takeoff rocket engines and rocket motors, and superperformance rocket engines for augmenting the thrust of an aircraft jet engine.

TABLE 1–4. Selected Military Applications Using Rocket Propulsion Systems

Category	Vehicle/System	Comments/Examples
Military Satellites	Reconnaissance/Observation Secure communications Early warning of ICBM launches	Rely on existing vehicles, such as Delta IV or Delta II and Atlas V
Strategic Weapons	ICBM—silo launched ICBM—submarine launched Cruise missile (subsonic flight) Intermediate range ballistic missile Battlefield support (very short range) Ship launched to ship or shore Small shoulder-fired missile Wire-guided small missile	Minuteman III; Trident; SPRMs; 5000 to 9000 km range; Tomahawk with SPRM booster and turbofan engine Pershing II (2 stages) Guided or unguided All use SPRM and single-stage, Redeye is a surface-to-air shoulder-fired missile
Surface-to-Air and Surface-to-Incoming-Missile Tactical weapons	Local area defense (e.g., airfield) Large area defense Battlefield support Shoulder-fired small missile Ship defense Short range Long range Carried under aircraft wings Antitank missiles Antisubmarine missiles Anti-radar missiles Torpedo propulsion Rocket-assisted artillery Aircraft pilot seat ejection	Standard missile, Patriot missile, Multistage vehicle with SPRM booster and LPRE divert top stage Guided or unguided SPRM, Redeye Unwinding wire to control flight path of local battlefield missile All use SPRMs Larger guided missiles Guided, SPRM, Phoenix, Sparrow With armor piercing warhead Hawk, TOW, SPRM Subroc, SPRM Homing air launched SPRM Gas generator, SPRM Up to 20,000g ₀ in gun barrel, increased range Emergency maneuver, SPRM

ICBM, intercontinental ballistic missile; LPRE, liquid propellant rocket engine; SPRM, solid propellant rocket motor.

TABLE 1-5. Selected Examples of Propulsion Characteristics of Rocket Applications

Application	Type of Propellant	Thrust Profile	Typical Firing Duration	Maximum Acceleration ^a
Large space launch vehicle	Liquid or cryogenic liquid	Nearly constant thrust	2-8 min	1.2-6 g ₀
Strap-on booster	Solid or liquid	100,000 to 3,300,000 lbf	½ to 2 min	1.2 to 3 g ₀
Spent strap-on stage separation	4 to 8 SPRMs	10,000 to 20,000 lbf each	Less than 1 sec	N. A.
Antiaircraft or antimissile-missile	Solid, some with liquid terminal divert stage	High thrust boost, decreasing thrust sustain phase; high thrust divert	2-75 sec each	5 to 20 g ₀ , but can be up to 100 g ₀
Spacecraft orbit maneuvers and/or maintenance	Storable liquid or cryogenic liquid; electric propulsion	Multiple restarts in space; can be pulsed	Up to 10 min	0.2-6 g ₀
Air-launched guided missile	Solid	High thrust boost phase with low thrust or decreasing thrust for sustain phase; sometimes 2 pulses	cumulative duration Boost: 2-5 sec Sustain: 10-30 sec	Up to 25 g ₀
Battlefield support—surface launched	Solid	Decreasing thrust	Up to 2 min each stage	Up to 10 g ₀
Rocket-assisted projectile, gun launched	Solid	Constant or decreasing thrust	A few sec	Up to 20,000 g ₀ in gun barrel
Spacecraft attitude control—large vehicles	Storable liquid (monopropellant or bipropellant); electric propulsion; xenon	Many restarts (up to several thousands); pulsing	Up to several hours	Less than 0.1 g ₀
Spacecraft attitude control—small vehicle	Electric propulsion; Cold or warm gas or storable liquid.	Same	Up to several hours	Same
Reusable main engines for Space Shuttle	Cryogenic liquid (O ₂ /H ₂)	Variable thrust, many flights with same engine	cumulative 8 min, over 7 hr cumulative in several missions	
Lunar landing	Storable bipropellant	10:1 thrust variation	4 min	Several g ₀
Weather sounding rocket	Solid	Single burn period—often decreasing thrust	5-30 sec	Up to 15 g ₀
Antitank	Solid	Single burn period	0.2-3 sec	Up to 20 g ₀

^a g₀ is acceleration of gravity at the Earth's surface = 9,8066 m/sec² or 32.17 ft/sec².

Other applications of rocket propulsion systems to space operations include communication satellites—these have been successfully deployed for many years providing relays of telephone and television signals between Earth stations; this application is managed and operated by commercial organizations. Other countries also have their own satellite-based communications. The U.S. government sponsored programs that include weather satellites and the now ubiquitous Global Positioning System (GPS). Examples of space exploration include missions to the planets and/or into deep space, such as the Voyager mission and the MESSENGER program. A good number of these missions and probes were developed by NASA's Jet Propulsion Laboratory; they use the multiple thrusters from a monopropellant hydrazine reaction control system. Another application is for suborbital winged space vehicles for space tourism; as of 2014, two such vehicles (privately financed) have been flown in experimental versions (see Fig. 16–3).

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CHAPTER 2

DEFINITIONS AND FUNDAMENTALS

This chapter deals with the definitions and basic relations for the propulsive force, exhaust velocity, and efficiencies related to creating and converting energy; comparisons of various propulsion systems and the simultaneous performance of multiple propulsion systems are also presented. The basic principles of rocket propulsion are essentially those of mechanics, thermodynamics, and chemistry. Propulsion is achieved by applying a force to a vehicle, that is, accelerating it or, alternatively, maintaining a given velocity against a resisting force. The propulsive force derives from momentum changes that originate from ejecting propellant at high velocities, and the equations in this chapter apply to all such systems. Symbols used in all equations are defined at the end of the chapter. Wherever possible, the American Standard letter symbols for rocket propulsion (as given in Ref. 2–1) are used.

2.1. DEFINITIONS

The *total impulse* I_t is found from the thrust force F (which may vary with time) integrated over the time of its application t :

$$I_t = \int_0^t F dt \quad (2-1)$$

For constant thrust with negligibly short start and stop transients, this reduces to

$$I_t = Ft \quad (2-2)$$

Total impulse I_t is essentially proportional to the total energy released by or into all the propellant utilized by the propulsion system.

The *specific impulse* I_s represents the thrust per unit propellant “weight” flow rate. It is an important figure of merit of the performance of any rocket propulsion system, a concept similar to miles per gallon parameter as applied to automobiles. A higher number often indicates better performance. Values of I_s are given in many chapters of this book and the concept of optimum specific impulse for a particular mission is introduced later. If the total propellant mass flow rate is $\dot{m}$ and the standard acceleration of gravity g_0 (with an *average* value at the Earth’s sea level of 9.8066 m/sec² or 32.174 ft/sec²), then

$$I_s = \frac{\int_0^t F dt}{g_0 \int_0^t \dot{m} dt} \quad (2-3)$$

This equation will give a time-averaged specific impulse value in units of “seconds” for any rocket propulsion system and is particularly useful when thrust varies with time. During transient conditions (during start or thrust buildup or shutdown periods, or during a change of flow or thrust levels) values of I_s may be obtained by either the integral above or by using average values for F and $\dot{m}$ for short time intervals. As written below, m_p represents the total effective propellant mass expelled.

$$I_s = I_t / (m_p g_0) \quad (2-4)$$

In Chapters 3, 12, and 17, we present further discussions of the specific impulse concept. For constant propellant mass flow $\dot{m}$, constant thrust F , and negligible start or stop transients Eq. 2-3 simplifies,

$$I_s = F / (\dot{m} g_0) = F / \dot{w} = I_t / w \quad (2-5)$$

At or near the Earth’s surface, the product $m_p g_0$ is the effective propellant weight w , and its corresponding weight flow rate given by $\dot{w}$. But for space or in satellite outer orbits, mass that has been multiplied by an “arbitrary constant,” namely, g_0 does not represent the weight. In the *Système International* (SI) or metric system of units, I_s is in “seconds.” In the United States today, we still use the English Engineering (EE) system of units (foot, pound, second) for many chemical propulsion engineering, manufacturing, and test descriptions. In many past and some current U.S. publications, data, and contracts, the specific impulse has units of thrust (lbf) divided by weight flow rate of the propellants (lbf/sec), also yielding the unit of seconds; thus, the numerical values for I_s are the same in the EE and the SI systems. Note, however, that this unit for I_s does not indicate a measure of elapsed time but the thrust force per unit “weight flow rate.” In this book, we use the symbol I_s exclusively for the specific impulse, as listed in Ref. 2-1. For solid propellants and other rocket propulsion systems, the symbol I_{sp} is more commonly used to represent specific impulse, as listed in Ref. 2-2.

In actual rocket nozzles, the exhaust velocity is not really uniform over the entire exit cross section and such velocity profiles are difficult to measure accurately. A uniform axial velocity c is assumed for all calculations which employ one-dimensional problem descriptions. This *effective exhaust velocity* c represents

an average or mass-equivalent velocity at which propellant is being ejected from the rocket vehicle. It is defined as

$$c = I_s g_0 = F / \dot{m} \quad (2-6)$$

It is given either in meters per second or feet per second. Since c and I_s only differ by a constant (g_0), either one can be used as a measure of rocket performance. In the Russian literature c is used in lieu of I_s .

In solid propellant rockets, it is difficult to measure propellant flow rate accurately. Therefore, in ground tests, the specific impulse is often calculated from total impulse and the propellant weight (using the difference between initial and final rocket motor weights and Eq. 2-5). In turn, the total impulse is obtained from the integral of the measured thrust with time, using Eq. 2-1. In liquid propellant units, it is possible to measure thrust and instantaneous propellant flow rate and thus Eq. 2-3 is used for the calculation of specific impulse. Equation 2-4 allows yet another interpretation for specific impulse, namely, the amount of total impulse imparted to a vehicle per total sea-level weight of propellant expended.

The term *specific propellant consumption* corresponds to the reciprocal of the specific impulse and is not commonly used in rocket propulsion. It is used in automotive and air-breathing duct propulsion systems. Typical values are listed in Table 1-2.

The *mass ratio* **MR** of the total vehicle or of a particular vehicle stage or of the propulsion system itself is defined to be the final mass m_f divided by the mass before rocket operation, m_0 . Here, m_f consists of the mass of the vehicle or stage after the rocket has ceased to operate when all the useful propellant mass m_p has been consumed and ejected. The various terms are depicted in Fig. 4-1.

$$\mathbf{MR} = m_f / m_0 \quad (2-7)$$

This equation applies to either a single or to a multistage vehicle; for the latter, the overall mass ratio is the product of the individual vehicle stage mass ratios. The final vehicle mass m_f has to include components such as guidance devices, navigation gear, the payload (e.g., scientific instruments or military warheads), flight control systems, communication devices, power supplies, tank structures, residual propellants, along with all the propulsion hardware. In some vehicles it may also include wings, fins, a crew, life support systems, reentry shields, landing gears, and the like. Typical values of **MR** can range from 60% for some tactical missiles down to 10% for some unmanned launch vehicle stages. This mass ratio is an important parameter for analyzing flight performance, as explained in Chapter 4. When the **MR** applies only to a single lower stage, then all its upper stages become part of its “payload.” It is important to specify when the **MR** applies either to a multiple-stage vehicle, to a single stage, or to a particular propulsion system.

The *propellant mass fraction* ζ indicates the ratio of the useful propellant mass m_p to the initial mass m_0 . It may apply to a vehicle, or a single stage, or to an entire rocket propulsion system.

$$\zeta = m_p/m_0 \quad (2-8)$$

$$\zeta = (m_0 - m_f)/m_0 = m_p/(m_p + m_f) \quad (2-9)$$

$$m_0 = m_p + m_f \quad (2-10)$$

Like the mass ratio **MR**, the propellant fraction ζ is used to describe a rocket propulsion system; its values will differ when applied to an entire vehicle, single or multistage. For a rocket propulsion system, the initial or loaded mass m_0 consists of the inert propulsion mass (the hardware necessary to burn and store the propellant) and the effective propellant mass. It would exclude masses of nonpropulsive components, such as payload or guidance devices (see Fig. 4–1). For example, in liquid propellant rocket engines the final or inert propulsion mass m_f would include propellant tanks, their feed and empty pressurization system (a turbopump and/or gas pressure system), one or more thrust chambers, various piping, fittings and valves, engine mounts or engine structures, filters, and some sensors. Any *residual or unusable remaining propellant* is normally considered to be part of the final inert mass m_f , as in this book. However, some rocket propulsion manufacturers and some literature assign residuals to be part of the propellant mass m_p . When applied to an entire rocket propulsion system, the value of the propellant mass fraction ζ indicates the quality of the design; a value of, say, 0.91 means that only 9% of the mass is inert rocket hardware, and this small fraction is needed to contain, feed, and burn the substantially larger mass of propellant; high values of ζ are desirable.

The *impulse-to-weight* ratio of a complete propulsion system is defined as the total impulse I_t divided by the initial (propellant-loaded) vehicle sea-level weight w_0 . A high value indicates an efficient design. Under our assumptions of constant thrust and negligible start and stop transients, it can be expressed as

$$\frac{I_t}{w_0} = \frac{I_t}{(m_f + m_p)g_0} = \frac{I_s}{m_f/m_p + 1} \quad (2-11)$$

The *thrust-to-weight* ratio F/w_0 expresses the acceleration (in multiples of the Earth's surface acceleration of gravity, g_0) that an engine is capable of giving to its own loaded propulsion system mass. Values of F/w_0 are given in Table 2–1. The *thrust-to-weight ratio* is useful in comparing different types of rocket propulsion systems and/or in identifying launch capabilities. For constant thrust the maximum value of the thrust-to-weight ratio, or maximum acceleration, invariably occurs just before thrust termination (i.e., burnout) because the vehicle's mass has been diminished by the mass of useful propellant.

TABLE 2-1. Ranges of Typical Performance Parameters for Various Rocket Propulsion Systems

Engine Type	Specific Impulse ^a (sec)	Maximum Temperature (°C)	Thrust-to-Weight Ratio ^b	Propulsion Duration	Specific Power ^c (kW/kg)	Typical Working Fluid	Status of Technology
Chemical—solid or liquid bipropellant, or hybrid	200–468	2500–4100	10^{-2} –100	Seconds to a few minutes	10^{-1} – 10^3	Liquid or solid propellants	Flight proven
Liquid monopropellant	194–223	600–800	10^{-1} – 10^{-2}	Seconds to minutes	0.02–200	N ₂ H ₄	Flight proven
Resistojet	150–300	2900	10^{-2} – 10^{-4}	Days	10^{-3} – 10^{-1}	H ₂ , N ₂ H ₄	Flight proven
Arc heating—electrothermal	280–800	20,000	10^{-4} – 10^{-2}	Days	10^{-3} –1	N ₂ H ₄ , H ₂ , NH ₃	Flight proven
Electromagnetic including pulsed plasma (PP)	700–2500	—	10^{-6} – 10^{-4}	Weeks	10^{-3} –1	H ₂	Flight proven
Hall effect	1220–2150	—	10^{-4}	Weeks	10^{-1} – 5×10^{-1}	Solid for PP Xenon	Flight proven
Ion—electrostatic	1310–7650	—	10^{-6} – 10^{-4}	Months, years	10^{-3} –1	Xenon	Flight proven
Solar heating	400–700	1300	10^{-3} – 10^{-2}	Days	10^{-2} –1	H ₂	In development

^aAt $p_1 = 1000$ psia and optimum gas expansion at sea level ($p_2 = p_3 = 14.7$ psia).

^bRatio of thrust force to full propulsion system sea level weight (with propellants, but without payload).

^cKinetic power per unit exhaust mass flow.

Example 2-1. A rocket vehicle has the following characteristics:

Initial mass	200 kg
Mass after rocket operation	130 kg
Payload, nonpropulsive structure, etc.	110 kg
Rocket operation duration	3.0 sec
Average specific impulse of propellant	240 sec

Determine the mass ratios for the vehicle and its propulsive unit, the propellant mass fraction for the vehicle and for its propulsive unit, the effective exhaust velocity, and the total impulse. Also, sensitive electronic equipment in the payload would limit the maximum acceleration to $35 g_0$'s. Is this exceeded during the flight? Assume constant thrust and neglect the short start and stop transients as well the force of gravity (which is being balanced by the lift of aerodynamic wings).

SOLUTION. The difference between initial mass and that after rocket operation is the mass of the expended propellant, namely, 70 kg. The mass ratio (Eq. 2-8) for the vehicle is therefore $\mathbf{MR} = m_f/m_0 = 130/200 = 0.65$, and for the propulsion unit it is $(130 - 110)/(200 - 110) = 0.222$; note that these are different. The propellant mass fraction of the propulsion system is (Eq. 2-8)

$$\zeta = 70/(200 - 110) = 0.778$$

This mass fraction is acceptable for tactical missiles but might be only "fair" for spacecraft. The effective exhaust velocity c is proportional to the specific impulse, Eq. 2-6, $c = 240 \times 9.81 = 2354.4 \text{ m/sec}$. The propellant mass flow rate is $\dot{m} = 70/3 = 23.3 \text{ kg/sec}$ and the thrust becomes (Eq. 2-6)

$$F = I_s g_0 = 23.3 \times 2354.4 = 54,936 \text{ N}$$

Now the total impulse can be calculated, $I_t = 54,936 \times 3 = 164,808 \text{ N-sec}$. The impulse-to-weight ratio (Eq. 2-11) for the propulsion system at the Earth's surface is $I_t/w_0 = 164,808/[(200 - 110)9.81] = 187 \text{ sec}$ (this might only reflect a "fair design" because a "good design" would approach a value closer to the specific impulse, $I_s = 240 \text{ sec}$).

For a horizontal trajectory, the maximum acceleration is found at the end of the thrusting schedule, just before shutdown (because while the thrust is unchanged the vehicle mass is now at its minimum value of 130 kg).

$$\text{Final acceleration} = F/m = 54,936/130 = 422.58 \text{ m/sec}^2$$

This value represents $43.08 g_0$'s, at the Earth's surface and it exceeds the stated $35 g_0$ limit of the equipment. This can be remedied by changing to a lower thrust along with a longer flight time.

2.2. THRUST

Thrust is the force produced by the rocket propulsion system acting at the vehicle's center of mass. It is a reaction force, experienced by vehicle's structure from the ejection of propellant at high velocities (the same phenomenon that pushes a garden

hose backward or makes a gun recoil). Momentum is a vector quantity defined as the product of mass times its vector velocity. In rocket propulsion, relatively small amounts of propellant mass *carried within* the vehicle are ejected at high velocities.

The thrust, due to a change in momentum, is shown below (a derivation can be found in earlier editions of this book). Here, the exit gas velocity is assumed constant, uniform, and purely axial, and when the mass flow rate is constant, thrust is itself constant. Often, this idealized thrust can actually be close to the actual thrust.

$$F = \frac{d(mv_2)}{dt} = \dot{m}v_2 \text{ at sea level} = \frac{\dot{w}}{g_0}v_2 \quad (2-12)$$

But this force represents the total propulsive force only when the nozzle exit pressure equals the ambient pressure.

The pressure of the surrounding fluid (i.e., the local atmosphere) gives rise to the second component of thrust. Figure 2-1 shows a schematic of an external uniform pressure acting on the outer surfaces of a rocket chamber as well as the changing gas pressures inside of a typical thermal rocket propulsion system. The direction and length of the arrows indicates the relative magnitude of the pressure forces. Axial thrust can be determined by integrating all the pressures acting on areas that have a projection on the plane normal to the nozzle axis. Any forces acting radially outward may be appreciable but do not contribute to the thrust when rocket chambers are axially symmetric.

For a fixed nozzle geometry, changes in ambient pressure due to variations in altitude during flight result in imbalances between the atmospheric pressure p_3 and the propellant gas pressure p_2 at the exit plane of the nozzle. For steady operation in a

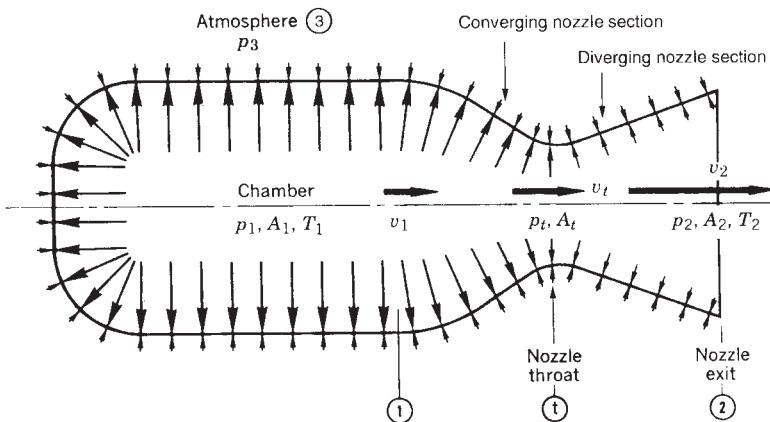


FIGURE 2-1. Gas pressures on the chamber and nozzle-interior walls are not uniform. The internal pressure (indicated by length of arrows) is highest in the chamber (p_1) and decreases steadily in the nozzle until it reaches the nozzle exit pressure p_2 . The external or atmospheric pressure p_3 is uniform. At the throat the pressure is p_t . The four subscripts (shown inside circles) are used to identify quantities such as A , v , T , and p at those specific locations. The centerline horizontal arrows denote relative velocities.

homogeneous atmosphere, the total thrust can be shown to equal (this equation is derived using the *control volume* approach in gas dynamics, see Refs. 2–3 and 2–4):

$$F = \dot{m}v_2 + (p_2 - p_3)A_2 \quad (2-13)$$

The first term is the *momentum thrust* given by the product of the propellant mass flow rate and its exhaust velocity relative to the vehicle. The second term represents the *pressure thrust*, consisting of the product of the cross-sectional area at the nozzle exit A_2 (where the exhaust jet leaves the vehicle) and the difference between the gas pressure at the exit and the ambient fluid pressure (p_2 may differ from p_3 only in supersonic nozzle exhausts). When the exit gas pressure is less than the surrounding fluid pressure, the pressure thrust is negative. Because this condition gives a lower thrust and is undesirable for other reasons (discussed in Chapter 3), rocket nozzles are usually designed so that their exhaust pressures equal or are slightly higher than the ambient pressure.

When the ambient or atmospheric pressure equals the exhaust pressure, the pressure term is zero and the thrust is the same as in Eq. 2–12. In the vacuum of space $p_3 = 0$ and the pressure thrust becomes a maximum,

$$F = \dot{m}v_2 + p_2A_2 \quad (2-14)$$

Most nozzles are have their area ratio A_2/A_t designed so that their exhaust pressure will equal the surrounding air pressure (i.e., $p_2 = p_3$) somewhere at or above sea level. For any fixed nozzle configuration this can occur only at one altitude, and this location is referred to as nozzle operation at its *optimum expansion ratio*; this case is treated in Chapter 3.

Equation 2–13 shows that thrust is independent of a rocket unit's flight velocity. Because changes in ambient pressure affect the pressure thrust, rocket thrust varies noticeably with altitude. Since atmospheric pressure decreases with increasing altitude, thrust and specific impulse will increase as the vehicle reaches higher altitudes. On Earth, this increase in pressure thrust due to altitude changes can amount to between 10 and 30% of the sea-level thrust; the fact that as flights gain altitude, atmospheric pressure continuously diminishes causing the thrust to increase is a unique feature of rocket propulsion systems. Table 8–1 shows the sea level and high-altitude thrust for several liquid rocket engines. Appendix 2 gives the properties of the *standard atmosphere* (the ambient pressure, p_3) as a function of altitude.

2.3. EXHAUST VELOCITY

The *effective exhaust velocity* as defined by Eq. 2–6 applies to all mass-expulsion thrusters. From Eq. 2–13 and for constant propellant mass flow it can be modified to give the equation below. As before, g_0 is a constant whose numerical value equals the average acceleration of gravity at sea level and does not vary with altitude.

$$c = v_2 + (p_2 - p_3)A_2/\dot{m} = I_s g_0 \quad (2-15)$$

In Eq. 2-6 the value of c may be determined from measurements of thrust and propellant flow. When $p_2 = p_3$, the value of the effective exhaust velocity c equals v_2 , the average actual nozzle exhaust velocity of the propellant gases. But even when $p_2 \neq p_3$ and $c \neq v_2$, the second term of the right-hand side of Eq. 2-15 usually remains small in relation to v_2 ; thus, the effective exhaust velocity always stays relatively close in value to the actual exhaust velocity. At the Earth's surface and when $c = v_2$, the thrust, from Eq. 2-13, can be simply written as (the second version below is less restricted)

$$F = (\dot{w}/g_0)v_2 = \dot{m}c \quad (2-16)$$

The *characteristic velocity* c^* , pronounced “cee-star,” is a term frequently used in the rocket propulsion literature. It is defined as

$$c^* = p_1 A_t / \dot{m} \quad (2-17)$$

The characteristic velocity c^* , though not a physical velocity, is used for comparing the relative performance of different chemical rocket propulsion system designs and propellants; it may be readily determined from measurements of $\dot{m}$, p_1 , and A_t . Being essentially independent of nozzle characteristics, c^* may also be related to the efficiency of the combustion process. However, the specific impulse I_s and the effective exhaust velocity c remain functions of nozzle geometry (such as the nozzle area ratio A_2/A_t , as shown in Chapter 3). Some typical values of I_s and c^* are given in Tables 5-5 and 5-6.

Example 2-2. The following measurements were made in a sea-level test of a solid propellant rocket motor (all cross sections are circular and unchanging):

Burn duration	40 sec
Initial propulsion system mass	1210 kg
Mass of rocket motor after test	215 kg
Sea-level thrust	62,250 N
Chamber pressure	7.00 MPa
Nozzle exit pressure	70.0 kPa
Nozzle throat diameter	8.55 cm
Nozzle exit diameter	27.03 cm

Determine $\dot{m}$, v_2 , c^* , and c at sea level. Also, determine the *pressure thrust* (Eq. 2-13) and the specific impulse at sea level, 1000 m, and 25,000 m altitude. Assume that the momentum thrust is invariant during the rocket ascent, and that start and stop transients can be neglected.

SOLUTION. The cross-sectional areas corresponding to the given diameters are $A_t = 0.00574$ and $A_2 = 0.0574 \text{ m}^2$ (i.e., an area ratio of 10). The steady-state mass flow rate for all altitudes is

$$\dot{m} = (1210 - 215)/40 = 24.88 \text{ kg/sec}$$

The desired c^* (all altitudes) and c at sea level follow from Eqs. 2-17 and 2-6:

$$c^* = p_1 A_t / \dot{m} = 7.00 \times 10^6 \times 0.00574 / 24.88 = 1615 \text{ m/sec}$$

$$c = F / \dot{m} = 62,250 / 24.88 = 2502 \text{ m/sec}$$

The *pressure thrust* at sea level is negative:

$$(p_2 - p_3)A_2 = (0.070 - 0.1013) \times 10^6 \times 0.0574 = -1797 \text{ N}$$

so the nozzle exit velocity v_2 becomes (for all altitudes), from Eq. 2-13,

$$v_2 = (62,250 + 1797)/24.88 = 2574 \text{ m/sec}$$

The remaining answers are shown in the table below. We obtain the ambient pressures from Appendix 2.

Altitude	p_3 (kPa)	Pressure Thrust (N)	I_s (sec)
Sea level	101.32	-1797	255
1000 m	89.88	-1141	258
25,000 m	2.55	3871	278

Further examination of the standard atmosphere table in Appendix 2 reveals that the *pressure thrust* becomes zero at just under 3000 m.

2.4. ENERGY AND EFFICIENCIES

Although efficiencies are not commonly used directly in designing rocket propulsion systems, they permit an understanding of the energy balance in these systems. Their definitions arbitrarily depend on the losses considered and any consistent set of efficiencies, such as the set presented in this section, is satisfactory in evaluating energy losses. As stated previously, two types of energy conversion processes occur in all propulsion systems, namely, the production of energy, which is most often the conversion of stored energy into available energy and, subsequently, its conversion to the form with which a reaction thrust can be obtained. The kinetic energy of ejected matter is the main useful form of energy for propulsion. The *power of the jet* P_{jet} is the time derivative of this energy, and for a constant gas exit velocity ($v_2 \approx c$ and at sea level), this is a function of I_s and F :

$$P_{\text{jet}} = \frac{1}{2} \dot{m} v_2^2 = \frac{1}{2} \dot{m} g_0 I_s^2 = \frac{1}{2} F g_0 I_s = \frac{1}{2} F v_2 \quad (2-18)$$

The term *specific power* is sometimes used as a measure of the utility of the mass of the propulsion system, including its power source; it equals the jet power divided by the loaded propulsion system mass, P_{jet}/m_0 . For electrical propulsion systems that need to include a massive, relatively inefficient energy source, specific power can be much lower than that for chemical rockets. The source of energy input to a rocket propulsion system is different in different thruster types; for chemical rockets this energy is created solely by combustion. The maximum energy available in chemical propellants is their heat of combustion per unit of propellant mass Q_R ; the *power input to a chemical engine* is given by

$$P_{\text{chem}} = \dot{m} Q_R J \quad (2-19)$$

where J is a units-conversion constant (see Appendix 1). A significant portion of the energy may leave the nozzle as residual enthalpy in the exhaust gases and is unavailable for conversion into kinetic energy. This is analogous to the energy lost in the hot exhaust gases of internal combustion engines.

The *combustion efficiency* η_{comb} for chemical rockets is the ratio of the actual energy released to the ideal heat of reaction per unit of propellant mass and represents a measure of the source efficiency. Its value can be high (nearly 94 to 99%). When the power input P_{chem} is multiplied by the combustion efficiency, it becomes the power available to the propulsive device, where it is then converted into the kinetic power of the exhaust jet. In electric propulsion the analogous efficiency is the power conversion efficiency; with solar cells this efficiency has a relatively low value because it depends on the efficiency of converting solar radiation energy into electric power (presently between 10 and 30%).

The *power transmitted to the vehicle* at any one instant of time is defined in terms of the product of the thrust of the propulsion system F and the *vehicle velocity* u :

$$P_{\text{vehicle}} = Fu \quad (2-20)$$

The *internal efficiency* of a rocket propulsion system reflects the effectiveness of converting the system's input energy to the propulsion device into kinetic energy of ejected matter; for example, for a chemical unit it is the ratio of the kinetic power of the ejected gases expressed by Eq. 2-18 divided by the power input of the chemical reaction, Eq. 2-19. The energy balance diagram for a chemical rocket (Fig. 2-2) shows typical losses. The internal efficiency η_{int} may be expressed as

$$\eta_{\text{int}} = \frac{\text{kinetic power in jet}}{\text{available chemical power}} = \frac{\frac{1}{2}\dot{m}v^2}{\eta_{\text{comb}}P_{\text{chem}}} \quad (2-21)$$

Any object moving through a fluid medium affects the fluid (i.e., stirs it) in ways that may hinder its motion and/or require extra energy expenditures. This is one consequence of skin friction, which can be substantial. The *propulsive efficiency* η_p (Fig. 2-3) reflects this energy cost for rocket vehicles. The equation that determines how much exhaust kinetic energy is useful for propelling a rocket vehicle is defined as

$$\begin{aligned} \eta_p &= \frac{\text{vehicle power}}{\text{vehicle power} + \text{residual kinetic jet power}} \\ &= \frac{Fu}{Fu + \frac{1}{2}\dot{m}(c-u)^2} = \frac{2u/c}{1 + (u/c)^2} \end{aligned} \quad (2-22)$$

where F is the thrust, u the absolute vehicle velocity, c the effective rocket exhaust velocity with respect to the vehicle, $\dot{m}$ the propellant mass flow rate, and η_p the desired propulsive efficiency. This propulsive efficiency becomes one when the forward vehicle velocity is exactly equal to the effective exhaust velocity; here any residual kinetic energy becomes zero and the exhaust gases effectively stand still in space.

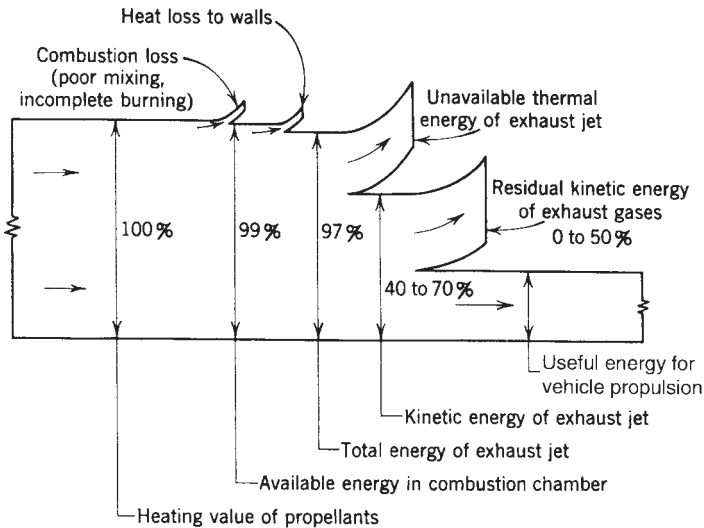


FIGURE 2-2. Typical energy distribution diagram for a chemical rocket.

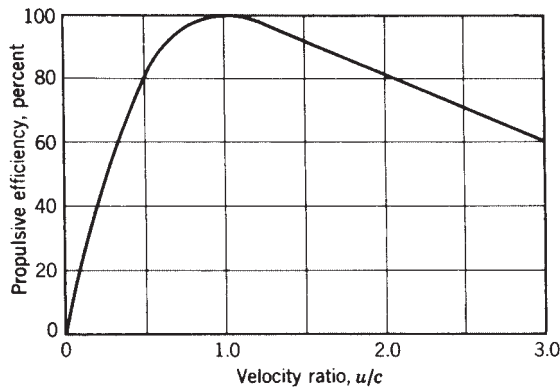


FIGURE 2-3. Propulsive efficiency at varying velocities.

While it is desirable to use energy economically and thus have high efficiencies, there is also the need for minimizing the expenditure of ejected mass, which in many cases is more important than minimizing the energy. With nuclear-reactor energy and some solar-energy sources, for example, there is an almost unlimited amount of heat energy available; yet the vehicle can carry only a limited amount of propellant. For a given thrust, economy of mass expenditures in the working fluid can be obtained when the exhaust velocity is high. Because the specific impulse is proportional to the exhaust velocity, it therefore measures this propellant mass economy.

2.5. MULTIPLE PROPULSION SYSTEMS

The relationships below are used for determining the total or overall (subscript oa) thrust and the overall mass flow of propellants for a group of propulsion systems (two or more) firing in parallel (i.e., in the same direction at the same time). These relationships apply to liquid propellant rocket engines, solid propellant rocket motors, electrical propulsion systems, hybrid propulsion systems, and any combinations of these. Many space launch vehicles and larger missiles use multiple propulsion systems. The Space Shuttle, for example, had three large liquid engines and two large solid motors firing jointly at liftoff.

Overall thrust, F_{oa} , is needed for determining the vehicle's flight path and overall mass flow rate, $\dot{m}_{oa}$, is needed for determining the vehicle's mass; together they determine the overall specific impulse, $(I_s)_{oa}$:

$$F_{oa} = \sum F = F_1 + F_2 + F_3 + \dots \quad (2-23)$$

$$\dot{m}_{oa} = \sum \dot{m} = \dot{m}_1 + \dot{m}_2 + \dot{m}_3 + \dots \quad (2-24)$$

$$(I_s)_{oa} = F_{oa}/(g_0 \dot{m}_{oa}) \quad (2-25)$$

For liquid propellant rocket engines with a turbopump and a gas generator, there is a separate turbine outlet flow that is usually dumped overboard through a pipe and a nozzle (see Fig. 1-4), which needs to be included in the equations above and this is treated in Examples 2-3 and 11-1.

Example 2-3. The MA-3 multiple liquid engine for the Atlas 2 missile had two booster engines ($F = 165,000$ lbf at sea level and $\dot{m} = 667$ lbm/sec each). The turbine exhaust gas of both boosters adds an extra 2300 lbf of thrust at a propellant flow rate of about 33 lbm/sec. These booster engines are dropped from the vehicle after about 145 sec operation. The central sustainer rocket engine, which also starts at liftoff, operates for a total 300 sec. It has a sea-level thrust of 57,000 lbf and altitude thrust of 70,000 lbf both at a mass flow of 270.5 lbm/sec. Its turbine exhaust gases are aspirated into the nozzle of the engine and do not directly contribute to the thrust. There are also two small vernier engines used for roll control of the vehicle during the sustainer-only portion of the flight. They each have an altitude thrust of 415 lbf with a propellant flow of 2.13 lbm/sec. Determine the overall thrust F_{oa} and the overall mass flow rate $\dot{m}_{oa}$ at liftoff and at altitude, when all the nozzles are pointing vertically down.

SOLUTION. Use Eqs. 2-23 and 2-24. The thrust from the turbine exhaust must be included in Eq. 2-23. At sea-level,

$$F_{oa} = 165,000 \times 2 + 2300 + 57,000 = 389,300 \text{ lbf}$$

$$\dot{m}_{oa} = 667 \times 2 + 33 + 270.5 = 1637.5 \text{ lbm/sec}$$

At altitude (vacuum),

$$F_{oa} = 70,000 + 2 \times 415 = 70,830 \text{ lbf}$$

$$\dot{m}_{oa} = 270.5 + 2 \times 2.13 = 274.76 \text{ lbm/sec}$$

2.6. TYPICAL PERFORMANCE VALUES

Typical values of representative performance parameters for different types of rocket propulsion are given in Table 2-1 and in Fig. 2-4.

Chemical rocket propulsion systems have relatively low values of specific impulse, relatively light machinery (i.e., low engine total mass), with very high thrust capabilities, and therefore can provide high acceleration and high specific power. At the other extreme, ion-propulsion thrusters have a very high specific impulse, but they carry massive electrical power systems to deliver the power necessary for such high ejection velocities. The very low acceleration potential for electrical propulsion units (and for others using solar energy) usually requires long accelerating periods and thus these systems are best suited for missions where powered flight times can be long. The low thrust magnitudes of electrical systems also imply that they are not useful in fields of strong gravitational gradients (e.g., for Earth takeoff or landing) but are best suited for flight missions in space.

Performance of rocket propulsion systems also depends on their application. Typical applications are shown in several chapters of this book (see Index).

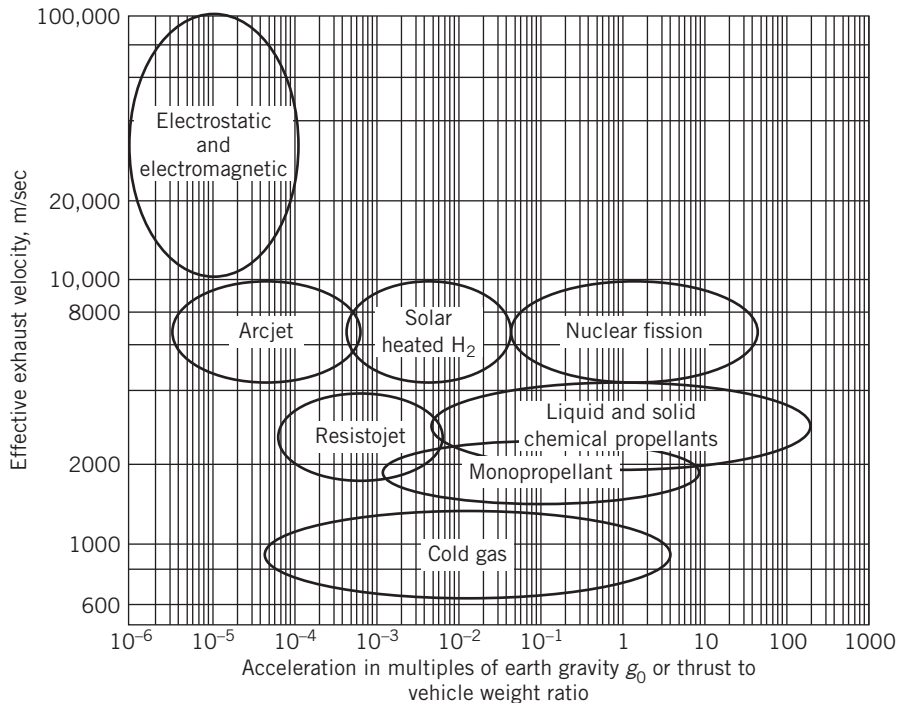


FIGURE 2-4. Exhaust velocities as a function of typical vehicle accelerations. Regions indicate approximate performance values for different types of propulsion systems. The mass of the vehicle includes the propulsion system, but the payload is assumed to be zero. Nuclear fission is shown only for comparison as it is not being pursued.

Chemical systems (solid and liquid propellant rockets) are presently the most developed and are widely used for many different vehicle applications. They are described in Chapters 5 to 16. Electrical propulsion has also been in operation in many space flight applications (see Chapter 17). Other types are still in their exploratory or development phase.

The accelerations shown in Figure 2–4 are directly related to the magnitudes of the thrust that is applied by the various rocket propulsion systems depicted. The effective exhaust velocities are related the required jet power per unit thrust (see Eq. 2–18) and the resulting input power (which is proportional to the square of the velocity and depends on the internal efficiency, Eq. 2–21) would turn out to be very high for electric propulsion systems unless the thrust itself is reduced to comparatively low values. As of 2015, electrical power levels of about 100 kW are expected to be available in space.

In Figure 2–4, hybrid rocket propulsion systems (see Chapter 16) are not shown separately because they are included as part of liquid and solid chemical propellants. Compressed (cold) gases stored at ambient temperatures have been used for many years for roll control in larger vehicles, for complete attitude control of smaller flight vehicles, and in rocket toys. Nuclear fission rocket propulsion systems as shown in Figure 2–4 represent analytical estimates; their development has been stopped. Hall and ion thrusters are part of the electrostatic and electromagnetic entry in this figure; they have flown successfully in many space applications (see Chapter 17).

2.7. VARIABLE THRUST

Most operational rocket propulsion systems have essentially constant propellant mass flow producing constant thrust or slightly increasing thrust with altitude. Only some flight missions require large thrust changes during flight; Table 2–2 shows several applications; the ones that require randomly variable thrust use predominantly liquid propellant rocket engines. Some applications require high thrust during a short initial period followed by a pre-programmed low thrust for the main flight portion (typically 20 to 35% of full thrust); these use predominantly solid propellant rocket motors. Section 8.8 describes how liquid propellant rocket engines can be designed and controlled for randomly variable thrust. Section 12.3 explains how the grain of solid rocket motors can be designed to give predetermined thrust changes.

Some solid and liquid propellant experimental propulsion systems have used variable nozzle throat areas (achieved with a variable position “tapered pintle” at the nozzle throat) and one experimental version has flown. To date, there has been no published information on production and implementation of such systems.

TABLE 2-2. Applications of Variable Thrust

Application	Type*	L/S*	Comment
1. Vertical ascent through atmosphere of large booster stage	AB	L	Reduced thrust avoids excessive aerodynamic pressure on vehicle
2. Short range tactical surface-to-surface missile	B	S	100 % initial thrust, 20 to 35% thrust for sustaining portion of flight
3. Tactical surface-to-air defensive missile	B	S	Same as # 2
4. Aircraft pilot emergency seat ejection capsule	B	S	Rapid ejection to get away from aircraft to go to higher altitude to deploy parachute
5. Soft landing on planet or moon, "retro-firing"	A	L	Thrust can be reduced by a factor of 10 with automatic landing controls
6. Top stage of multistage area defense missile against attacking ballistic missile	A or B	L, S	Axial thrust, side thrust, and attitude control thrust to home in on predicted vehicle impact point
7. Sounding rocket or weather rocket (vertical ascent)	B	S	Programmed two-thrust levels for many, but not all such rockets

*A Random variable thrust.

*B Preprogrammed (decreasing) thrust profile.

*L Liquid propellant rocket engine.

*S Solid propellant rocket motor.

SYMBOLS

A	area, m^2 (ft^2)
A_t	nozzle throat area, m^2 (ft^2)
A_2	exit area of nozzle, m^2 (ft^2)
c	effective exhaust velocity, m/sec (ft/sec)
c^*	characteristic velocity, m/sec (ft/sec)
E	energy, J (ft-lbf)
F	thrust force, N (lbf)
F_{oa}	overall force, N (lbf)
g_0	average sea-level acceleration of gravity, 9.81 m/sec^2 (32.2 ft/sec^2), [at equator 9.781 , at poles 9.833 m/sec^2]
I_s	specific impulse, sec
$(I_s)_{oa}$	overall specific impulse, sec
I_t	impulse or total impulse, N-sec (lbf-sec)
J	conversion factor or mechanical equivalent of heat, 4.184 J/cal or 1055 J/Btu or 778 ft-lbf/Btu

m	mass, kg (slugs, 1 slug = mass of a 32.174 lb-weight at sea level)
m_{oa}	overall mass, kg
$\dot{m}$	mass flow rate, kg/sec (lbm/sec)
m_f	final mass (after rocket propellant is ejected), kg (lbm or slugs)
m_p	propellant mass, kg (lbm or slugs)
m_0	initial mass (before rocket propellant is ejected), kg (lbm or slugs)
MR	mass ratio (m_f/m_0)
p	pressure, pascal [Pa] or N/m ² (lbf/ft ²)
p_3	ambient or atmospheric pressure, Pa (lbf/ft ²)
p_2	rocket gas pressure at nozzle exit, Pa (lbf/ft ²)
p_1	chamber pressure, Pa (lbf/ft ²)
P	power, J/sec (ft-lbf/sec)
P_s	specific power, J/sec-kg (ft-lbf/sec-lbm)
Q_R	heat of reaction per unit propellant, J/kg (Btu/lbm)
t	time, sec
u	vehicle velocity, m/sec (ft/sec)
v_2	gas velocity leaving the nozzle, m/sec (ft/sec)
w	weight, N or kg-m/sec ² (lbf)
$\dot{w}$	weight flow rate, N/sec (lbf/sec)
w_0	initial weight, N or kg-m/sec ² (lbf)

Greek Letters

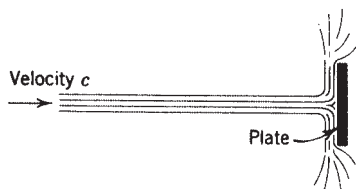
ζ	propellant mass fraction
η	efficiency
η_{comb}	combustion efficiency
η_{int}	internal efficiency
η_p	propulsive efficiency

PROBLEMS

When solving problems, three appendixes (see end of book) may be helpful:

- Appendix 1. Conversion Factors and Constants
- Appendix 2. Properties of the Earth's Standard Atmosphere
- Appendix 3. Summary of Key Equations

1. A jet of water hits a stationary flat plate in the manner shown below.



- a. If 50 kg per minute flows at an absolute velocity of 200 m/sec, what will be the force on the plate?
- b. What will this force be when the plate moves in the flow direction at $u = 50$ km/h? Explain your methodology.

Answers: 167 N; 144 N.

2. The following data are given for a certain rocket unit: thrust, 8896 N; propellant consumption, 3.867 kg/sec; velocity of vehicle, 400 m/sec; energy content of propellant, 6.911 MJ/kg. Assume 100% combustion efficiency.

Determine (a) the effective velocity; (b) the kinetic jet energy rate per unit flow of propellant; (c) the internal efficiency; (d) the propulsive efficiency; (e) the overall efficiency; (f) the specific impulse; (g) the specific propellant consumption.

Answers: (a) 2300 m/sec; (b) 2.645 MJ/kg; (c) 38.3%; (d) 33.7%; (e) 13.3%; (f) 234.7 sec; (g) 0.00426 sec^{-1} .

3. A certain rocket engine (flying horizontally) has an effective exhaust velocity of 7000 ft/sec; it consumes 280 lbm/sec of propellant mass, and liberates 2400 Btu/lbm. The unit operates for 65 sec. Construct a set of curves plotting the propulsive, internal, and overall efficiencies versus the velocity ratio u/c ($0 < u/c < 1.0$). The rated flight velocity equals 5000 ft/sec. Calculate (a) the specific impulse; (b) the total impulse; (c) the mass of propellants required; (d) the volume that the propellants occupy if their average specific gravity is 0.925. Neglect gravity and drag.

Answer: (a) 217.4 sec; (b) 3,960,000 lbf-sec; (c) 18,200 lbm; (d) 315 ft^3 .

4. For the rocket in Problem 2, calculate the specific power, assuming a propulsion system dry mass of 80 kg and a duration of 3 min.
5. A Russian rocket engine (RD-110 with LOX-kerosene) consists of four thrust chambers supplied by a single turbopump. The exhaust from the turbine of the turbopump then is ducted to four vernier nozzles (which can be rotated to provide some control of the flight path). Using the information below, determine the thrust and mass flow rate of the four vernier gas nozzles. For individual thrust chambers (vacuum):

$$F = 73.14 \text{ kN}, c = 2857 \text{ m/sec}$$

For overall engine with verniers (vacuum):

$$F = 297.93 \text{ kN}, c = 2845 \text{ m/sec}$$

Answers: 5.37 kN, 2.32 kg/sec.

6. A certain rocket engine has a specific impulse of 250 sec. What range of vehicle velocities (u , in units of ft/sec) would keep the propulsive efficiencies at or greater than 80%. Also, how could rocket-vehicle staging be used to maintain these high propulsive efficiencies for the range of vehicle velocities encountered during launch?

Answers: 4021 to 16,085 ft/sec; design upper stages with increasing I_s .

7. For a solid propellant rocket motor with a sea-level thrust of 207,000 lbf, determine: (a) the (constant) propellant mass flow rate $\dot{m}$ and the specific impulse I_s at sea level, (b) the altitude for optimum nozzle expansion as well as the thrust and specific impulse at this

optimum condition and (c) at vacuum conditions. The initial total mass of the rocket motor is 50,000 lbm and its propellant mass fraction is 0.90. The residual propellant (called slivers, combustion stops when the chamber pressure falls below a deflagration limit) amounts to 3 % of the burnt. The burn time is 50 seconds; the nozzle throat area (A_t) is 164.2 in.² and its area ratio (A_2/A_t) is 10. The chamber pressure (p_1) is 780 psia and the pressure ratio (p_1/p_2) across the nozzle may be taken as 90.0. Neglect any start/stop transients and use the information in Appendix 2.

Answers: (a) $\dot{m} = 873$ lbm/sec, 237 sec., (b) $F = 216,900$ lbf, $I_s = 248.5$ sec., (c) $F = 231,000$ lbf, $I_s = 295$ sec.

8. During the boost phase of the Atlas V, the RD-180 engine operates together with three solid propellant rocket motors (SRBs) for the initial stage. For the remaining thrust time, the RD-180 operates alone. Using the information given in Table 1–3, calculate the *overall effective exhaust velocity* for the vehicle during the initial combined thrust operation.

Answer: 309 sec.

9. Using the values given in Table 2–1, choose three propulsion systems and calculate the total impulse for a fixed propellant mass of 20 kg.
10. Using the MA-3 rocket engine information given in Example 2–3, calculate the overall specific impulse at sea level and at altitude, and compare these with I_s values for the individual booster engines, the sustainer engine, and the individual vernier engines.

Answers: (I_s)_{oa} = 238 sec (SL) and 258 sec (altitude)

11. Determine the mass ratio **MR** and the mass of propellant used to produce thrust for a solid propellant rocket motor that has an inert mass of 82.0 kg. The motor mass becomes 824.5 kg after loading the propellant. For safety reasons, the igniter is not installed until shortly before motor operation; this igniter has a mass of 5.50 kg of which 3.50 kg is igniter propellant. Upon inspection after firing, the motor is found to have some unburned residual propellant and a motor mass of 106.0 kg.

Answers: **MR** = 0.1277, propellant burned = 720.5 kg.

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CHAPTER 3

NOZZLE THEORY AND THERMODYNAMIC RELATIONS

In rocket propulsion systems the mathematical tools needed to calculate performance and to determine several key design parameters involve the principles from gas dynamics and thermodynamics that describe processes inside a rocket nozzle and its chamber. These relations are also used for evaluating and comparing the performance between different rocket systems since with them we can predict operating parameters for any system that uses the thermodynamic gas expansion in a supersonic nozzle; they allow the determination of nozzle size and generic shape for any given performance requirement. This theory applies to chemical rocket propulsion systems (liquid and solid and hybrid propellant types), nuclear rockets, solar-heated and resistance or arc-heated electrical rocket systems, and all propulsion systems that use gas expansion as the mechanism for ejecting matter at high velocities.

Fundamental thermodynamic relations are introduced and explained in this chapter. By using these equations, the reader can gain a basic understanding of the thermodynamic processes involved in high-temperature and/or pressure gas expansions. Some knowledge of both elementary thermodynamics and fluid mechanics on the part of the reader is assumed (see Refs. 3–1 to 3–3). This chapter also addresses different nozzle configurations, nonoptimum performance, energy losses, nozzle alignment, variable thrust, and four different alternate nozzle performance parameters.

3.1. IDEAL ROCKET PROPULSION SYSTEMS

The concept of an ideal rocket propulsion system is useful because the relevant basic thermodynamic principles can be expressed with relatively simple mathematical

relationships, as shown in subsequent sections of this chapter. These equations describe quasi-one-dimensional nozzle flows, which represent an idealization and simplification of the full two- or three-dimensional equations of real aerothermochemical behavior. However, within the assumptions stated below, these descriptions are very adequate for obtaining useful solutions to many rocket propulsion systems and for preliminary design tasks. In chemical rocket propulsion, measured actual performances turn out to be usually between 1 and 6% below the calculated ideal values. In designing new rocket propulsion systems, it has become accepted practice to use such ideal rocket parameters, which can then be modified by appropriate corrections, such as those discussed in Section 3.5. An *ideal rocket* propulsion unit is defined as one for which the following assumptions are valid:

1. The working fluid (which usually consists of chemical reaction products) is *homogeneous in composition*.
2. All the species of the working fluid are treated as *gaseous*. Any condensed phases (liquid or solid) add a negligible amount to the total mass.
3. The working fluid obeys the *perfect gas law*.
4. There is *no heat transfer* across any and all gas-enclosure walls; therefore, the flow is *adiabatic*.
5. There is no appreciable *wall friction* and all *boundary layer* effects may be neglected.
6. There are no *shock waves* or other *discontinuities* within the nozzle flow.
7. The *propellant flow rate* is *steady* and *constant*. The expansion of the working fluid is uniform and steady, without gas pulsations or significant turbulence.
8. *Transient effects* (i.e., start-up and shutdown) are of such short duration that they may be neglected.
9. All exhaust gases leaving the rocket nozzles travel with a *velocity parallel to the nozzle axis*.
10. The gas velocity, pressure, temperature, and density are all *uniform* across any section normal to the nozzle axis.
11. *Chemical equilibrium* is established within the preceding combustion chamber and gas composition does not change in the nozzle (i.e., frozen composition flow).
12. Ordinary propellants are stored at ambient temperatures. Cryogenic propellants are at their boiling points.

These assumptions permit the derivation of the relatively compact, quasi-one-dimensional set of equations shown in this chapter. Later in this book we present more sophisticated theories and/or introduce correction factors for several of the items on the above list, which then allow for more accurate determinations. The next paragraph explains why the above assumptions normally cause only small errors.

For liquid bipropellant rockets, the idealized situation postulates an injection system in which the fuel and oxidizer mix perfectly so that a homogeneous working

medium results; a good rocket injector can closely approach this condition. For solid propellant rocket units, the propellant must essentially be homogeneous and uniform and the burning rate must be steady. For solar-heated or arc-heated propulsion systems, it must be assumed that the hot gases can attain a uniform temperature at any cross section and that the flow is steady. Because chamber temperatures are typically high (2500 to 3600 K for common propellants), all gases are well above their respective saturation conditions and do follow closely the perfect gas law. Assumptions 4, 5, and 6 above allow the use of *isentropic expansion* relations within the rocket nozzle, thereby describing the maximum conversion from heat and pressure to kinetic energy of the jet (this also implies that the nozzle flow is thermodynamically reversible). Wall friction losses are difficult to determine accurately, but they are usually negligible when the inside walls are smooth. Except for very small chambers, the heat losses to the walls of the rocket are usually less than 1% (occasionally up to 2%) of the total energy and can therefore be neglected. Short-term fluctuations in propellant flow rates and pressures are typically less than 5% of their steady value, small enough to be neglected. In well-designed supersonic nozzles, the conversion of thermal and/or pressure energy into directed kinetic energy of the exhaust gases may proceed smoothly and without normal shocks or discontinuities—thus, flow expansion losses are generally small.

Some rocket companies and/or some authors do not include all or the same 12 items listed above in their definition of an ideal rocket. For example, instead of assumption 9 (all nozzle exit velocity is axially directed), they use a conical exit nozzle with a 15° half-angle as their base configuration for the ideal nozzle; this item accounts for the divergence losses, a topic later described in this chapter (using the correction factor λ).

3.2. SUMMARY OF THERMODYNAMIC RELATIONS

In this section we briefly review some of the basic relationships needed for the development of the nozzle flow equations. Rigorous derivations and discussions of these relations can be found in many thermodynamics or fluid dynamics texts, such as Refs. 3–1 to 3–3.

The principle of *conservation of energy* may be readily applied to the adiabatic, no shaft-work processes inside the nozzle. Furthermore, in the absence of shocks or friction, flow entropy changes are zero. The concept of *enthalpy* is most useful in flow systems; the enthalpy h comprises the *internal thermal energy* plus the *flow work* (or work performed by the gas of velocity v in crossing a boundary). For ideal gases the enthalpy can conveniently be expressed as the product of the specific heat c_p times the absolute temperature T (c_p is the specific heat at constant pressure, defined as the partial derivative of the enthalpy with respect to temperature at constant pressure). Under the above assumptions, the total or stagnation enthalpy per unit mass h_0 remains constant in nozzle flows, that is,

$$h_0 = h + v^2/2J = \text{constant} \quad (3-1)$$

Other stagnation conditions are introduced later, below Eq. 3–7. The symbol J is the mechanical equivalent of heat which is utilized only when thermal units (i.e., the Btu and calorie) are mixed with mechanical units (i.e., the ft-lbf and the joule). In SI units (kg, m, sec) the value of J is one. In the EE (English Engineering) system of units the value of the constant J is given in Appendix 1. Conservation of energy applied to isentropic flows between any two nozzle axial sections x and y shows that the decrease in static enthalpy (or thermodynamic content of the flow) appears as an increase of kinetic energy since any changes in potential energy may be neglected.

$$h_x - h_y = \frac{1}{2}(v_y^2 - v_x^2)/J = c_p(T_x - T_y) \quad (3-2)$$

The principle of *conservatism of mass* in steady flow, for passages with a single inlet and single outlet, is expressed by equating the mass flow rate $\dot{m}$ at any section x to that at any other section y ; this is known as the mathematical form of the continuity equation. Written in terms of the cross-sectional area A , the velocity v , and the “specific volume” V (i.e., the volume divided by the mass within), at any section

$$\dot{m}_x = \dot{m}_y \equiv \dot{m} = Av/V \quad (3-3)$$

The *perfect gas law* can be written as (at an arbitrary location x)

$$p_x V_x = RT_x \quad (3-4)$$

where the gas constant R is found from the universal gas constant R' divided by the molecular mass $\mathfrak{M}$ of the flowing gas mixture. The molecular volume at standard conditions becomes $22.41 \text{ m}^3/\text{kg} - \text{mol}$ or $\text{ft}^3/\text{lb} - \text{mol}$ and it relates to a value of $R' = 8314.3 \text{ J/kg-mol-K}$ or $1544 \text{ ft-lbf/lb-mol-}^\circ\text{R}$. We often find Eq. 3–3 written in terms of density ρ which is the reciprocal of the specific volume V . The specific heat at constant pressure c_p , the specific heat at constant volume c_v , and their ratio k are constant for perfect gases over a wide range of temperatures and are related as follows:

$$k = c_p/c_v \quad (3-5a)$$

$$c_p - c_v = R/J \quad (3-5b)$$

$$c_p = kR/(k - 1)J \quad (3-6)$$

For any *isentropic flow process*, the following relations may be shown to hold between any two nozzle sections x and y :

$$T_x/T_y = (p_x/p_y)^{(k-1)/k} = (V_y/V_x)^{k-1} \quad (3-7)$$

During an isentropic expansion the pressure drops substantially, the absolute temperature drops somewhat less, and the specific volume increases. When flows are stopped isentropically the prevailing conditions are known as *stagnation conditions* which are designated by the subscript 0. Sometimes the word “total” is used instead of

stagnation. As can be seen from Eq. 3-1 the stagnation enthalpy consists of the sum of the static or local enthalpy and the fluid kinetic energy. The absolute stagnation temperature T_0 is found from this energy equation as

$$T_0 = T + v^2/(2c_p J) \quad (3-8)$$

where T is the static absolute fluid temperature. In adiabatic flows, the stagnation temperature remains constant. A useful relationship between the stagnation pressure and the local pressure in isentropic flows can be found from the previous two equations:

$$p_0/p = [1 + v^2/(2c_p J T)]^{k/(k-1)} = (V/V_0)^k \quad (3-9)$$

When local velocities are close to zero, the corresponding local temperatures and pressures approach the stagnation pressure and stagnation temperature. Inside combustion chambers, where gas velocities are typically small, the local combustion pressure essentially equals the stagnation pressure. Now, the *velocity of sound* a also known as the acoustic velocity in perfect gases is independent of pressure. It is defined as

$$a = \sqrt{kRT} \quad (3-10)$$

In the EE system the units of R must be corrected and a conversion constant $g_c \equiv g_0$ must be added – Equation 3-10 becomes $\sqrt{g_c kRT}$. This correction factor allows for the proper velocity units. The *Mach number* M is a dimensionless flow parameter and is used to locally define the ratio of the flow velocity v to the local acoustic velocity a :

$$M = v/a = v/\sqrt{kRT} \quad (3-11)$$

Hence, Mach numbers less than one correspond to subsonic flows and greater than one to supersonic flows. Flows moving at precisely the velocity of sound would have Mach numbers equal to one. It is shown later that at the throat of all one-dimensional supersonic nozzles the Mach number must be equal to one. The relation between stagnation temperature and Mach number may now be written from Eqs. 3-2, 3-7, and 3-10 as

$$T_0 = T \left[1 + \frac{1}{2}(k-1)M^2 \right] \quad (3-12)$$

or

$$M = \sqrt{\frac{2}{k-1} \left(\frac{T_0}{T} - 1 \right)}$$

T_0 and p_0 designate the temperature and pressure stagnation values. Unlike the temperature, the stagnation pressure during an adiabatic nozzle expansion only remains constant for totally isentropic flows (i.e., no losses of any kind). It may be computed from

$$p_0 = p \left[1 + \frac{1}{2}(k-1)M^2 \right]^{k/(k-1)} \quad (3-13)$$

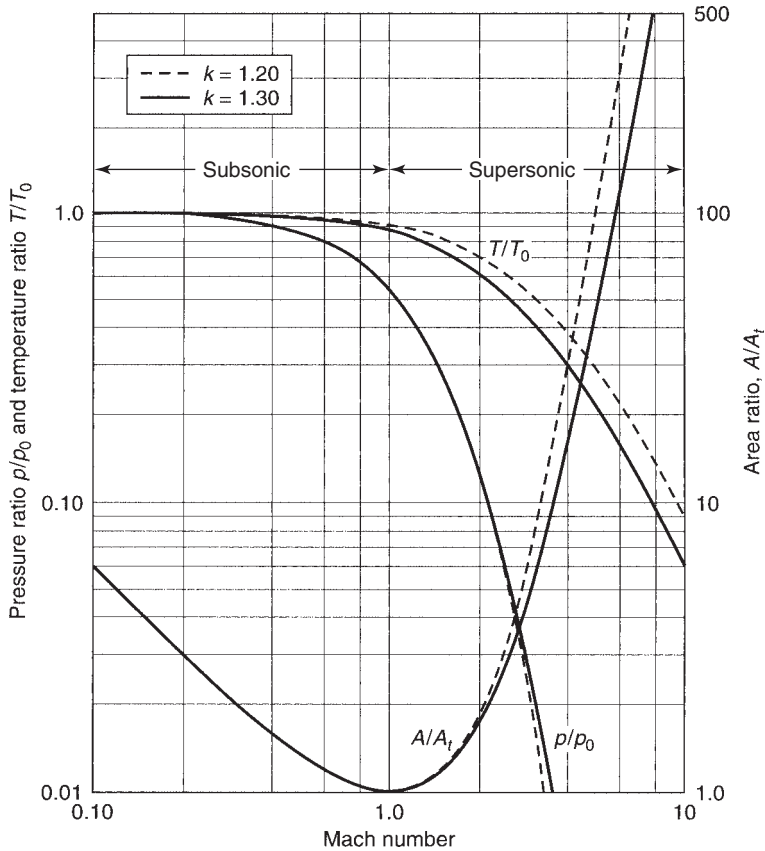


FIGURE 3–1. Relationship of area ratio, pressure ratio, and temperature ratio as functions of Mach number in a converging/diverging nozzle depicted for the subsonic and supersonic nozzle regions.

The nozzle *area ratio* for isentropic flow may now be expressed in terms of Mach numbers for two arbitrary locations x and y within the nozzle. Such a relationship is plotted in Fig. 3–1 for $M_x = 1.0$, where $A_x = A_t$ the throat or minimum area, along with corresponding ratios for T/T_0 and p/p_0 . In general,

$$\frac{A_y}{A_x} = \frac{M_x}{M_y} \sqrt{\left\{ \frac{1 + [(k-1)/2]M_y^2}{1 + [(k-1)/2]M_x^2} \right\}^{(k+1)/(k-1)}} \quad (3-14)$$

As can be seen from Fig. 3–1, for subsonic flows any chamber contraction (from station $y = 1$) or nozzle entrance ratio A_1/A_t can remain small, with values approaching 3 to 6 depending on flow Mach number, and the passage is convergent. There are no noticeable effects from variations of k . In solid rocket motors the chamber area

A_1 refers to the flow passage or port cavity in the virgin grain. With supersonic flow the nozzle section diverges and the area ratio enlarges rather quickly; area ratios are significantly influenced by the value of k . At $M = 4$ the exit area ratio A_2/A_t ranges between 15 and 30, depending on the value of k indicated. On the other hand, pressure ratios are less sensitive to k but temperature ratios show more variation.

The average *molecular mass* $\mathfrak{M}$ of a mixture of gases is the sum of all chemical species' n_i in kg-mols multiplied by their molecular mass ($n_i\mathfrak{M}_i$) and then divided by the sum of all molar masses. This is further discussed in Chapter 5, Eq. 5–5. The symbol $\mathfrak{M}$ has been used to avoid confusion with M for the Mach number. In the older literature $\mathfrak{M}$ is called the molecular weight.

Example 3–1. An “ideal rocket engine” is designed to operate at sea level using a propellant whose products of combustion have a specific heat ratio k of 1.3. Determine the required chamber pressure if the exit Mach number is 2.52. Also determine the nozzle area ratio between the throat and exit.

SOLUTION. For “optimum expansion” the nozzle exit pressure must equal the local atmospheric pressure, namely, 0.1013 MPa. If the chamber velocity may be neglected, then the ideal chamber pressure is the total stagnation pressure, which can be found from Eq. 3–13 as

$$\begin{aligned} p_0 &= p \left[1 + \frac{1}{2}(k-1)M^2 \right]^{k/(k-1)} \\ &= 0.1013 [1 + 0.15 \times 2.52^2]^{1.3/0.3} = 1.84 \text{ MPa} \end{aligned}$$

The ideal nozzle area ratio A_2/A_t is determined from Eq. 3–14 setting $M_t = 1.0$ at the throat (see also Fig. 3–1):

$$\begin{aligned} \frac{A_2}{A_t} &= \frac{1}{M_2} \left[\frac{1 + [(k-1)/2]M_2^2}{(k+1)/2} \right]^{(k+1)/2(k-1)} \\ &= \frac{1}{2.52} \left[\frac{1 + 0.15 \times 2.52^2}{1.15} \right]^{2.3/0.6} = 3.02 \end{aligned}$$

Note that *ideal* implies no losses, whereas *optimum* is a separate concept reflecting the best calculated performance at a particular set of given pressures. *Optimum* performance is often taken as the design condition and it occurs when $p_2 = p_3$ as will be shown in the section on the *thrust coefficient* (the peak of the curves in Figs. 3–6 and 3–7 for fixed p_1/p_3).

3.3. ISENTROPIC FLOW THROUGH NOZZLES

Within converging–diverging nozzles a large fraction of the thermal energy of the gases flowing from the chamber is converted into kinetic energy. As will be shown, gas pressures and temperatures may drop dramatically and gas velocities can reach values in excess of 2 miles per second. This expansion is taken as a reversible or isentropic flow process in the analyses described here. When a nozzle’s inner wall has a flow obstruction or a wall protrusion (a piece of weld splatter or slag), then the

gas kinetic energy is locally converted back into thermal energy essentially recovering the stagnation temperature and stagnation pressure in the chamber. Since this would quickly lead to local material overheating and wall failure, all nozzle inner walls must be smooth without any protrusions. Stagnation conditions can also occur at the leading edge of a jet vane (described in Chapter 18) or at the tip of a gas sampling tube inserted into the flow.

Velocity

The nozzle exit velocity (at $y = 2$) v_2 can be solved for from Eq. 3-2:

$$v_2 = \sqrt{2J(h_1 - h_2) + v_1^2} \quad (3-15a)$$

As stated, this relation strictly applies when there are no heat losses. This equation also holds between any two locations within the nozzle, but hereafter subscripts 1 and 2 will only designate nozzle inlet and exit conditions. For constant k the above expression may be rewritten with the aid of Eqs. 3-6 and 3-7:

$$v_2 = \sqrt{\frac{2k}{k-1} RT_1 \left[1 - \left(\frac{p_2}{p_1} \right)^{(k-1)/k} \right] + v_1^2} \quad (3-15b)$$

When the chamber cross section is large compared the nozzle throat, the chamber velocity or nozzle entrance velocity is comparatively small and the term v_1^2 may be neglected above. The chamber temperature T_1 , which is located at the nozzle inlet, under isentropic conditions differs little from the stagnation temperature or (for chemical rocket propulsion) from the combustion temperature T_0 . Hence we arrive at important equivalent simplified expressions of the velocity v_2 , ones commonly used for analysis:

$$\begin{aligned} v_2 &= \sqrt{\frac{2k}{k-1} RT_1 \left[1 - \left(\frac{p_2}{p_1} \right)^{(k-1)/k} \right]} \\ &= \sqrt{\frac{2k}{k-1} \frac{R'T_0}{\mathfrak{M}} \left[1 - \left(\frac{p_2}{p_1} \right)^{(k-1)/k} \right]} \end{aligned} \quad (3-16)$$

As can be seen above, the gas velocity exhausting from an ideal nozzle is a function of the prevailing *pressure ratio* p_1/p_2 , the ratio of specific heats k , and the absolute temperature at the nozzle inlet T_1 , as well as of the gas constant R . Because the gas constant for any particular gas is inversely proportional to the molecular mass $\mathfrak{M}$, exhaust velocities or their corresponding specific impulses strongly depend on the ratio of the absolute nozzle entrance temperature (which is close to T_0) divided by the average molecular mass of the exhaust gases, as is shown in Fig. 3-2. The fraction

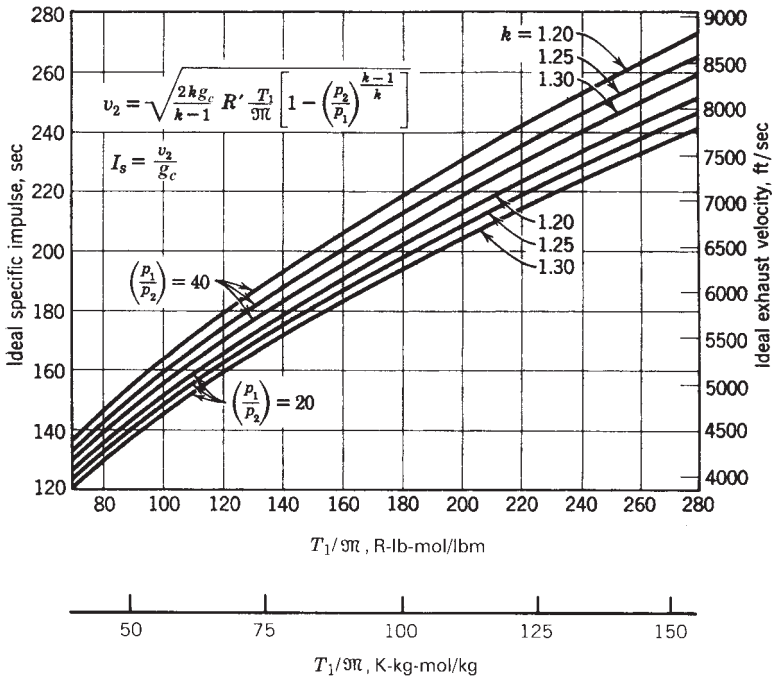


FIGURE 3-2. Specific impulse and exhaust velocity of an ideal rocket propulsion unit at optimum nozzle expansion as functions of the absolute chamber gas temperature T_1 and the molecular mass M for several values of k and p_1/p_2 .

T_0/M plays an important role in optimizing the mixture ratio (oxidizer to fuel flow) in chemical rocket propulsion defined in Eq. 6-1.

Equations 2-13 and 2-16 give relations between the thrust F and the velocities v_2 and c , and the corresponding specific impulse I_s which is plotted in Fig. 3-2 for two pressure ratios and three values of k . Equation 3-16 clearly shows that any increase in entrance gas temperatures (arising from increases in chamber energy releases) and/or any decrease of propellant molecular mass (commonly achieved from using low molecular mass gas mixtures rich in hydrogen) will enhance the ratio T_0/M ; that is, they will increase the specific impulse I_s through increases in the exhaust velocity v_2 and, thus, the performance of the rocket vehicle. Influences of pressure ratio across the nozzle p_1/p_2 and of specific heat ratio k are less pronounced. As can be seen from Fig. 3-2, performance does increase with an increase of the pressure ratio; this ratio increases when the value of the chamber pressure p_1 rises and/or when the exit pressure p_2 decreases, corresponding to high altitude designs. The small influence of k values is fortuitous because low molecular masses, found in many diatomic and monatomic gases, correspond to the higher values of k .

Values of the pressure ratio must be standardized when comparing specific impulses from one rocket propulsion system to another or for evaluating the influence of various design parameters. Presently, a chamber pressure of 1000

psia (6.894 MPa) and an exit pressure of 1 atm (0.1013 MPa) are used as standard reference for such comparisons, or $p_1/p_2 = 68.06$.

For *optimum expansion* $p_2 = p_3$ and the effective exhaust velocity c (Eq. 2-15) and the ideal rocket exhaust velocity v_2 become equal, namely,

$$v_2 = (c)_{\text{opt}} \quad (3-17)$$

Thus, here c can be substituted for v_2 in Eqs. 3-15 and 3-16. For a fixed nozzle exit area ratio, and constant chamber pressure, this optimum condition occurs only at the particular altitude where the ambient pressure p_3 just equals the nozzle exhaust pressure p_2 . At all other altitudes $c \neq v_2$.

The maximum theoretical value of nozzle outlet velocity is reached with an infinite nozzle expansion (when exhausting into a vacuum):

$$(v_2)_{\text{max}} = \sqrt{2kRT_0/(k-1)} \quad (3-18)$$

This maximum theoretical exhaust velocity is finite, even though the pressure ratio is not, because it corresponds to finite thermal energy contents in the fluid. In reality, such an expansion cannot occur because, among other things, the temperature of most gas species will eventually fall below their liquefaction or their freezing points; thus, they cease to be a gas and can no longer contribute to the expansion or to any further velocity increases.

Example 3-2. A rocket propulsion system operates near sea level with a chamber pressure of $p_1 = 2.068$ MPa or 300 psia, a chamber temperature of 2222 K, and a propellant consumption of $\dot{m} = 1.0$ kg/sec. Take $k = 1.30$ and $R = 345.7$ J/kg - K. Calculate the ideal thrust and the ideal specific impulse. Also plot the cross-sectional area A , the local velocity v , the specific volume V , the absolute temperature T , and the local Mach number M with respect to pressure along the nozzle.

SOLUTION. Assume that the operation of this rocket propulsion system is at an optimum for expansion to sea-level pressure $p_3 = 0.1013$ MPa so that $p_2/p_1 = 0.049$. In. From Eq. 3-16 the effective exhaust velocity is calculated to be 1827 m/sec since here $v_2 = c$. Hence,

$$I_s = c/g_0 = 1827/9.81 = 186 \text{ sec} \quad \text{from Eq. 2-5} \quad \text{and}$$

$$F = \dot{m}c = 1.0 \times 1827 = 1827 \text{ N} \quad \text{from Eq. 2-16}$$

In this example, we denote the nozzle axial location x as either the nozzle entrance (location 1 in Fig. 2-1) condition or the throat section (location t in Fig. 2-1) and keep the axial location y as a variable which spans the nozzle ranging from 1 $\rightarrow$ 2 (exit) in Fig. 2-1.

The calculation of the velocity follows from Eq. 3-16 with p_y set equal to the exit pressure p_3 , the local atmospheric pressure or 0.1013 MPa.

$$\begin{aligned} v_2 &= \sqrt{\frac{2k}{k-1} RT_1 \left[1 - \left(\frac{p_y}{p_1} \right)^{(k-1)/k} \right]} \\ &= \sqrt{\frac{2 \times 1.30 \times 345.7 \times 2222}{1.30 - 1} \left[1 - \left(\frac{0.1013}{2.068} \right)^{0.2308} \right]} = 1827 \text{ m/sec} \end{aligned}$$

The cross-sectional area can be obtained from Eq. 3–3 and the Mach number from Eq. 3–11. By stepwise varying the pressure from 2.068 to 0.1013 MPa, one can arrive at the information shown in Fig. 3–3; it is a tedious process. A faster way is to modify the relevant equations in terms of p_y so that they can be plotted directly and this is shown below.

The initial specific volume V_1 is calculated from the equation of state of a perfect gas (Eq. 3–4),

$$V_1 = RT_1/p_1 = 345.7 \times 2222/(2.068 \times 10^6) = 0.3714 \text{ m}^3/\text{kg}$$

Now we can write equations for the specific volume and the temperature (Eq. 3–7) as a function of the pressure p_y

$$V_y = V_1(p_1/p_y)^{1/k} = 0.3714(2.068/p_y)^{0.769} \text{ (m}^3/\text{kg)}$$

$$T_y = T_1(p_y/p_1)^{(k-1)/k} = 2222(p_y/2.068)^{0.231} \text{ (K)}$$

These can be plotted by inserting p_y values between 2.068 and 0.1013 in MPa. The calculations for the velocity follow from Eq. 3–16,

$$v_y = 2580[1 - (p_y/2.068)^{0.231}]^{1/2} \text{ (m/sec)}$$

The cross-sectional area is found from Eq. 3–3 and the relations above:

$$\begin{aligned} A_y &= \dot{m} V_y / v_y = (1.0 \times 0.3714 / 2580) \times (2.068/p_y)^{0.769} [1 - (p_y/2.068)^{0.231}]^{-1/2} \\ &= 1.44 \times 10^{-4} \times (2.068/p_y)^{0.769} [1 - (p_y/2.068)^{0.231}]^{-1/2} \text{ (m}^2\text{)} \end{aligned}$$

Finally the Mach number is obtained using Eq. 3–11 and the relations above:

$$\begin{aligned} M_y &= v_y / [kRT_y]^{1/2} \\ &= [2580 / (1.30 \times 345.7 \times 2222)^{1/2}] [1 - (p_y/2.068)^{0.231}]^{1/2} (p_y/2.068)^{-0.231} \\ &= 2.582 [1 - (p_y/2.068)^{0.231}]^{1/2} (p_y/2.068)^{-0.231} \end{aligned}$$

With the equations indicated above and graphing software such as MATLAB or MAPLE or other, the desired plots may be obtained. Figure 3–3 shows semiquantitative plots of key parameter variations and should only be used to exhibit trends. Note that in this figure the pressures are shown as decreasing to the right, representing a nozzle flow from left to right (also its units are in psia where the throat pressure is 164.4 psia and the exit pressure would be 14.7 psia, just to the left of zero).

A number of interesting deductions can be made from this example. Very *high gas velocities* (over 1 km/sec) can be obtained in rocket nozzles. The *temperature drop* of the combustion gases flowing through a rocket nozzle can be appreciable. In the example above the temperature changed 1115 °C in a relatively short distance.

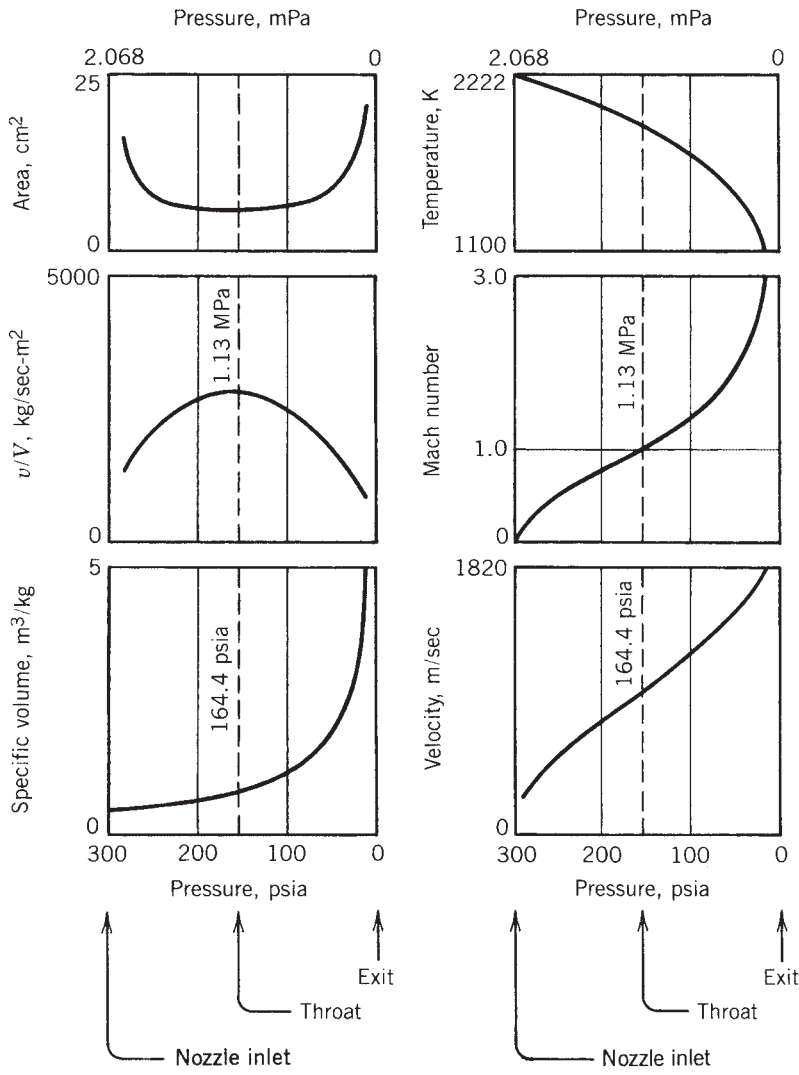


FIGURE 3-3. Typical variations of cross-sectional area, temperature, specific volume, Mach number, and velocity with pressure in a rocket nozzle (from Example 3-2).

This should not be surprising, for the increase in the kinetic energy of the gases comes from a decrease of the enthalpy, which in turn is roughly proportional to the decrease in temperature. Because the exhaust gases are still relatively hot (1107 K) when leaving the nozzle, they contain considerable thermal energy which is not available for conversion because it is seldom realistic to match exit and ambient temperatures, (see Fig. 2-2).

Nozzle Flow and Throat Condition

Supersonic nozzles (often called De Laval nozzles after their inventor) always consist of a convergent section leading to a *minimum* area followed by a divergent section. From the continuity equation, this area is inversely proportional to the ratio v/V . This quantity has also been shown in Fig. 3–3, where there is a *maximum* in the curve of v/V because at first the velocity increases at a greater rate than the specific volume; however, in the divergent section, the specific volume increases faster.

The minimum nozzle cross-sectional area is commonly referred to as the *throat area* A_t . The ratio of the nozzle exit area A_2 to the throat area A_t is called the *nozzle expansion area ratio* and is designated here by the Greek letter ϵ . It represents an important nozzle parameter:

$$\epsilon = A_2/A_t \quad (3-19)$$

The maximum gas flow rate per unit area occurs at the throat where a unique gas pressure ratio exists, which is only a function of the ratio of specific heats k . This pressure ratio is found by setting $M = 1$ in Eq. 3–13:

$$p_t/p_1 = [2/(k+1)]^{k/(k-1)} \quad (3-20)$$

The throat pressure p_t for which an isentropic mass flow rate attains its maximum is called the *critical pressure*. Typical values of the above critical pressure ratio range between 0.53 and 0.57 of the nozzle inlet pressure. The flow through a specified rocket nozzle with a given inlet condition is less than the maximum if the pressure ratio is larger than that given by Eq. 3–20. However, note that this ratio is not the value across the entire nozzle and that the maximum flow or choking condition (explained below) is always established internally at the throat and not at the exit plane. The nozzle inlet pressure is usually very close to the chamber stagnation pressure, except in narrow combustion chambers where there is an appreciable drop in pressure from the injector region to the nozzle entrance region. This is discussed in Section 3.5. At the location of critical pressure, namely the throat, the Mach number is always one and the values of the specific volume and temperature can be obtained from Eqs. 3–7 and 3–12:

$$V_t = V_1[(k+1)/2]^{1/(k-1)} \quad (3-21)$$

$$T_t = 2T_1/(k+1) \quad (3-22)$$

In Eq. 3–22 the nozzle inlet temperature T_1 is typically very close to the combustion chamber temperature and hence close to the nozzle flow stagnation temperature T_0 . At the throat there can only be a small variation for these properties. Take, for example, a gas with $k = 1.2$; the critical pressure ratio is about 0.56 (which means that p_t is a little more than half of the chamber pressure p_1 ; the temperature drops only slightly ($T_t = 0.91T_1$), and the specific volume expands by over 60% ($V_t = 1.61V_1$). Now, from Eqs. 3–15, 3–20, and 3–22, the *critical or throat velocity* v_t is obtained:

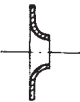
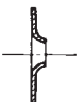
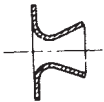
$$v_t = \sqrt{\frac{2k}{k+1} RT_1} = \sqrt{kRT_t} = a_t \quad (3-23)$$

The first version of this equation permits the critical velocity to be calculated directly from the nozzle inlet conditions and the second one gives v_t when throat temperature is known. The throat is the only nozzle section where the flow velocity is also the *local sonic velocity* ($a_t = v_t$ because $M_t = 1.00$). The inlet flow from the chamber is subsonic and downstream of the nozzle throat it is supersonic. The divergent portion of the nozzle provides for further decreases in pressure and increases in velocity under supersonic conditions. If the nozzle is cut off at the throat section, the exit gas velocity is sonic and the flow rate per unit area will remain its maximum. Sonic and supersonic flow conditions can only be attained when the critical pressure prevails at the throat, that is, when p_2/p_1 is equal to or less than the quantity defined by Eq. 3-20. There are, therefore, three basically different types of nozzles: subsonic, sonic, and supersonic, and these are described in Table 3-1.

The supersonic nozzle is the only one of interest for rocket propulsion systems. It achieves a high degree of conversion of enthalpy to kinetic energy. The ratio between the inlet and exit pressures in all rocket propulsion systems must be designed sufficiently large to induce supersonic flow. Only when the absolute chamber pressure drops below approximately 1.78 atm will there be subsonic flow in the divergent portion of any nozzle during sea-level operation. This pressure condition actually occurs for very short times during start and stop transients.

The velocity of sound, a , is the propagation speed of an elastic pressure wave within the medium, sound being an infinitesimal pressure wave. If, therefore, sonic velocity is reached at any location within a steady flow system, it is impossible for pressure disturbances to travel upstream past that location. Thus, any partial obstruction or disturbance of the flow downstream of a nozzle throat running critical has no influence on the throat or upstream of it, provided that the disturbance does not raise the downstream pressure above its critical value. It is not possible to increase the sonic velocity at the throat or the flow rate in a fixed configuration nozzle by further lowering the exit pressure or even evacuating the exhaust section. This important condition has been described as *choking* the flow, which is always established at the throat (and not at the nozzle exit or any other plane). *Choked mass flow* rates through the critical section of a supersonic nozzle may be derived from Eqs. 3-3, 3-21, and 3-23.

TABLE 3-1. Nozzle Types

	Subsonic	Sonic	Supersonic
Throat velocity	$v_1 < a_t$	$v_t = a_t$	$v_t = a_t$
Exit velocity	$v_2 < a_2$	$v_2 = v_t$	$v_2 > v_t$
Mach number	$M_2 < 1$	$M_2 = M_t = 1.0$	$M_2 > 1$
Pressure ratio	$\frac{p_1}{p_2} < \left(\frac{k+1}{2}\right)^{k/(k-1)}$	$\frac{p_1}{p_2} = \frac{p_1}{p_t} = \left(\frac{k+1}{2}\right)^{k/(k-1)}$	$\frac{p_1}{p_2} > \left(\frac{k+1}{2}\right)^{k/(k-1)}$
Shape			

From continuity, $\dot{m}$ always equals to the mass flow at any other cross section within the nozzle for steady flow.

$$\dot{m} = \frac{A_t v_t}{V_t} = A_t p_1 k \frac{\sqrt{[2/(k+1)]^{(k+1)/(k-1)}}}{\sqrt{kRT_1}} \quad (3-24)$$

The (choked) mass flow through a rocket nozzle is therefore proportional to the throat area A_t and the chamber (stagnation) pressure p_1 ; it is also inversely proportional to the square root of $T/\mathfrak{M}$ as well as a function of other gas properties. For supersonic nozzles the *ratio between the throat and any downstream area* at which a pressure p_y prevails can be expressed as a function of pressure ratios and the ratio of specific heats, by using Eqs. 3-4, 3-16, 3-21, and 3-23, as follows:

$$\frac{A_t}{A_y} = \frac{V_t v_y}{V_y v_t} = \left(\frac{k+1}{2}\right)^{1/(k-1)} \left(\frac{p_y}{p_1}\right)^{1/k} \sqrt{\frac{k+1}{k-1} \left[1 - \left(\frac{p_y}{p_1}\right)^{(k-1)/k}\right]} \quad (3-25)$$

When $p_y = p_2$, then $A_y/A_t = A_2/A_t = \epsilon$ and Eq. 3-25 shows the inverse nozzle exit area expansion ratio. For low-altitude operation (sea level to about 10,000 m) nozzle area ratios are typically between 3 and 30, depending on chamber pressure, propellant combinations, and vehicle envelope constraints. For high altitudes (100 km or higher) area ratios are typically between 40 and 200, but there have been some as high as 400. Similarly, an expression for the ratio of the velocity at any point downstream of the throat with a pressure p_y to the throat velocity may be written from Eqs. 3-15 and 3-23 as:

$$\frac{v_y}{v_t} = \sqrt{\frac{k+1}{k-1} \left[1 - \left(\frac{p_y}{p_1}\right)^{(k-1)/k}\right]} \quad (3-26)$$

Note that Eq. 3-26 is dimensionless but does *not* represent a Mach number. Equations 3-25 and 3-26 permit direct determination of the velocity ratio or the area ratio for any given pressure ratio, and vice versa, in ideal rocket nozzles. They both are plotted in Fig. 3-4. When $p_y = p_2$, Eq. 3-26 describes the velocity ratio between the nozzle exit area and the throat section. When the exit pressure coincides with the atmospheric pressure ($p_2 = p_3$, see Fig. 2-1), these equations apply to optimum nozzle expansions. For rockets that operate at high altitudes, it can be shown that not much additional exhaust velocity may be gained by increasing the area ratio above 1000. In addition, design difficulties and a heavy inert nozzle mass have made applications above area ratios of about 350 marginal.

Appendix 2 is a table of several properties of the Earth's atmosphere with established standard values. It gives ambient pressure for different altitudes. These properties do vary somewhat from day to day (primarily because of solar activity) and between hemispheres. For example, the density of the atmosphere at altitudes between 200 and 3000 km can change by more than an order of magnitude, affecting satellite drag estimates.

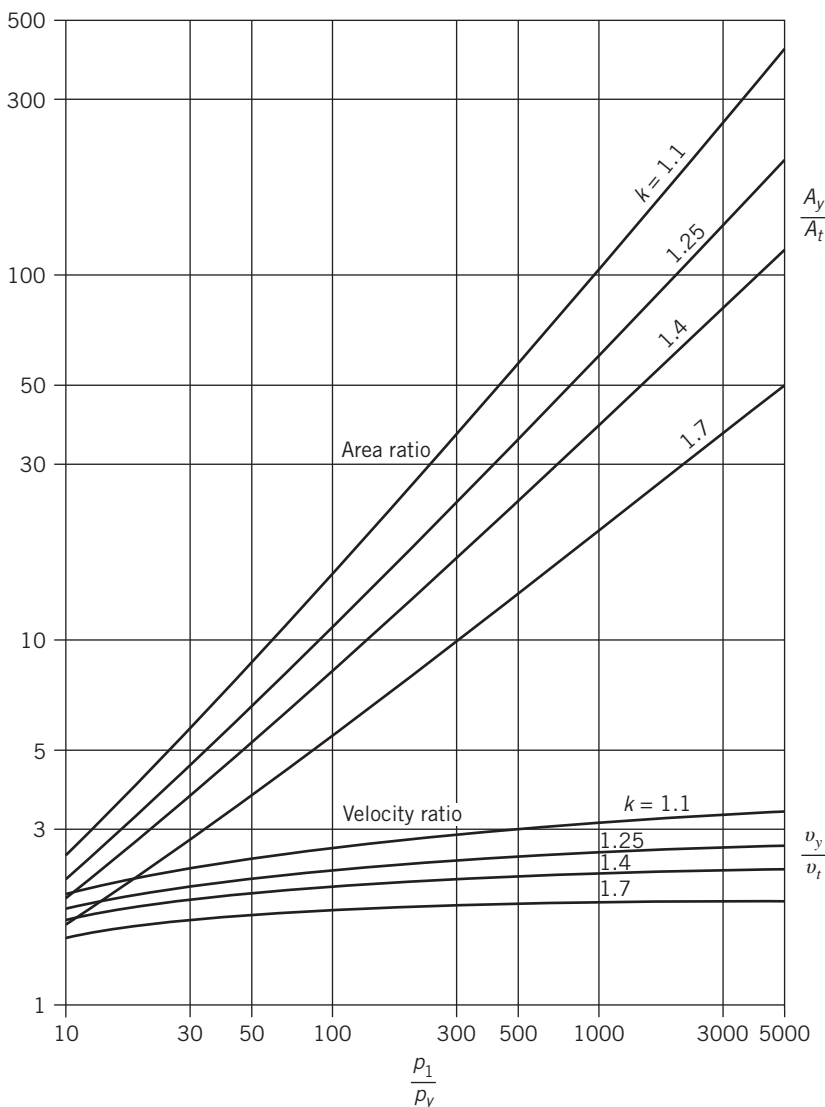


FIGURE 3-4. Area and velocity ratios as function of pressure ratio for the diverging section of a supersonic nozzle.

Example 3-3. Design an ideal nozzle for a rocket propulsion system that operates at 25 km altitude and delivers 5000 N thrust with a chamber pressure of 2.039 MPa and a chamber temperature of 2800 K. For this particular propellant, $k = 1.20$ and $R = 360$ J/kg-K. Determine the throat area, exit area, throat temperature, and exit velocity for optimum expansion.

SOLUTION. At 25 km altitude the atmospheric pressure equals 2.549 kPa (in Appendix 2 the ratio shown is 0.025158, which must be multiplied by the pressure at sea level, 0.1013 MPa).

The controlling pressure ratio is

$$p_2/p_1 = 0.002549/2.039 = 0.00125 = 1/800$$

The area ratio may now be obtained directly from Eq. 3-25 (or Fig. 3-4) as

$$\begin{aligned} \frac{A_t}{A_2} &= \left(\frac{k+1}{2} \right)^{1/(k-1)} \left(\frac{p_2}{p_1} \right)^{1/k} \sqrt{\frac{k+1}{k-1} \left[1 - \left(\frac{p_2}{p_1} \right)^{(k-1)/k} \right]} \\ &= (1.10)^5 (0.00125)^{0.833} \sqrt{11.0 [1 - (0.00125)^{0.167}]} \\ &= 1.67 \times 10^{-2} = 1/60 \end{aligned}$$

The temperature at the throat may be found directly from Eq. 3-22:

$$T_t = 2T_1/(k+1) = 2 \times 2800/2.2 = 2545 \text{ K}$$

The ideal exit velocity is found from Eq. 3-16 as

$$\begin{aligned} v_2 &= \sqrt{\frac{2k}{k-1} RT_1 \left[1 - \left(\frac{p_2}{p_1} \right)^{(k-1)/k} \right]} \\ &= \sqrt{1.21 \times 10^7 [1 - (0.00125)^{0.167}]} = 2851 \text{ m/sec} \end{aligned}$$

Next, we need to calculate the throat area, which can be found via the mass flow rate, Eq. 3-24, $A_t = \dot{m} V_t/v_t$. To arrive at the mass flow rate explicitly we note that $v_2 = c$ because $p_2 = p_3$ (Eq. 2-15) and from Eq. 2-16

$$\begin{aligned} \dot{m} &= F/c = 5000/2851 = 1.754 \text{ kg/sec} \\ A_t &= \frac{\dot{m}}{p_1} \sqrt{\frac{RT_1}{k[2/(k+1)]^{(k+1)/(k-1)}}} \\ &= \frac{1.754}{2.039 \times 10^6} \sqrt{\frac{360 \times 2800}{1.2[2/2.2]^{11}}} = 13.32 \text{ cm}^2 \end{aligned}$$

And thus the ideal exit area becomes $A_2 = 60 \times 13.32 = 799 \text{ cm}^2$. The designer would next have to choose an actual nozzle configuration and modify the area ratio to account for nonideal effects, and to choose materials together with cooling methods that can accommodate the high chamber and throat temperatures.

Thrust and Thrust Coefficient

As already stated, the efflux of propellant gases (i.e., their momentum flowing out) causes thrust—a reaction force on the rocket structure. Because this flow is supersonic, the pressure at the exit plane of the nozzle may differ from the ambient pressure, and a pressure component in the thrust equation adds (or subtracts) to the momentum thrust as given by Eq. 2-13, which is repeated here:

$$F = \dot{m}v_2 + (p_2 - p_3)A_2$$

Maximum thrust for any given nozzle operation is found in a vacuum where $p_3 = 0$. Between sea level and the vacuum of space, Eq. 2-13 gives the variation of thrust with altitude, applying the properties of the atmosphere as listed in Appendix 2. A typical variation of thrust with altitude from a choked nozzle would be represented by a smooth increase curving to an asymptotic value (see Example 2-2). To modify values calculated for the optimum operating condition ($p_2 = p_3$), for other conditions of p_1 , k , and ϵ , the following expressions may be used. For the thrust,

$$F = F_{\text{opt}} + p_1 A_t \left(\frac{p_2}{p_1} - \frac{p_3}{p_1} \right) \epsilon \quad (3-27)$$

For the specific impulse, using Eqs. 2-5, 2-17, and 2-13,

$$I_s = (I_s)_{\text{opt}} + \frac{c^* \epsilon}{g_0} \left(\frac{p_2}{p_1} - \frac{p_3}{p_1} \right) \quad (3-28)$$

When, for example, the specific impulse at a new exit pressure p_2 corresponding to a new area ratio A_2/A_t is to be calculated, the above relations may be used.

Equation 2-13 can be modified by substituting v_2 , v_t , and V_t from Eqs. 3-16, 3-21, and 3-23:

$$\begin{aligned} F &= \frac{A_t v_t v_2}{V_t} + (p_2 - p_3) A_2 \\ &= A_t p_1 \sqrt{\frac{2k^2}{k-1} \left(\frac{2}{k+1} \right)^{(k+1)/(k-1)} \left[1 - \left(\frac{p_2}{p_1} \right)^{(k-1)/k} \right]} + (p_2 - p_3) A_2 \quad (3-29) \end{aligned}$$

The first version of this equation is general and applies to the gas expansion of all rocket propulsion systems, the second form applies to an ideal rocket propulsion system with k being constant throughout the expansion process. This equation shows that thrust is proportional to the throat area A_t and the chamber pressure (or the nozzle inlet pressure) p_1 and is a function of the pressure ratio across the nozzle p_1/p_2 , the specific heat ratio k , and of the pressure thrust. Equation 3-29 is called the ideal thrust equation. A thrust coefficient C_F may now be defined as the thrust divided by the chamber pressure p_1 and the throat area A_t . Equation 3-30 then results:

$$\begin{aligned} C_F &= \frac{F}{p_1 A_t} = \frac{v_2^2 A_2}{p_1 A_t V_2} + \frac{p_2 A_2}{p_1 A_t} - \frac{p_3 A_2}{p_1 A_t} \\ &= \sqrt{\frac{2k^2}{k-1} \left(\frac{2}{k+1} \right)^{(k+1)/(k-1)} \left[1 - \left(\frac{p_2}{p_1} \right)^{(k-1)/k} \right]} + \frac{p_2 - p_3}{p_1} \frac{A_2}{A_t} \quad (3-30) \end{aligned}$$

The thrust coefficient C_F is dimensionless. It is a key parameter for analysis and depends on the gas property k , the nozzle area ratio ϵ , and the pressure ratio across the nozzle p_1/p_2 , but is not directly dependent on chamber temperature. For any fixed

ratio p_1/p_3 , the thrust coefficient C_F and the thrust F have a peak when $p_2 = p_3$ as area ratio $\epsilon = A_2/A_t$ varies. This peak value is known as the *optimum thrust coefficient* and is an important criterion in nozzle designs. The introduction of the thrust coefficient permits the following form of Eq. 3–29:

$$F = C_F A_t p_1 \quad (3-31)$$

Equation 3–31 may be solved for C_F to experimentally determine thrust coefficient from measured values of chamber pressure, throat diameter, and thrust. Even though the thrust coefficient is a function of chamber pressure, it is not simply proportional to p_1 , as can be seen from Eq. 3–30. However, it is directly proportional to throat area. Thrust coefficient has values ranging from just under 1.0 to roughly 2.0. Thrust coefficient may be thought of as representing the amplification of thrust due to the gas expansion in the supersonic nozzle as compared to the thrust that would be exerted if the chamber pressure acted over the throat area only. It is a convenient parameter for visualizing effects of chamber pressure and/or altitude variations in a given nozzle configuration, or for modifying sea-level results to flight altitude conditions.

Figure 3–5 shows variations of the *optimum expansion thrust coefficient* ($p_2 = p_3$) with pressure ratio p_1/p_2 , for different values of k and area ratio ϵ . Complete thrust coefficient variations for $k = 1.20$ and 1.30 are plotted in Figs. 3–6 and 3–7 as a function of pressure ratio p_1/p_3 and area ratio ϵ . These sets of curves are very useful in examining various nozzle problems for they permit the evaluation of under- and overexpanded nozzle operation, as explained below. The values given in these figures are ideal and do not consider such losses as divergence, friction, or internal expansion waves.

When p_1/p_3 becomes very large (e.g., expansions into near vacuum), the thrust coefficient approaches an asymptotic maximum as shown for $k = 1.20$ and 1.30 in Figs. 3–6 and 3–7. These figures also give values of C_F for some mismatched nozzle conditions ($p_2 \neq p_3$), provided the nozzle is flowing full at all times (the working fluid does not separate or break away from the nozzle walls). Flow separation is discussed later in this section.

Characteristic Velocity and Specific Impulse

The *characteristic velocity* c^* was defined by Eq. 2–17. From Eqs. 3–24 and 3–31, it can be shown that

$$c^* = \frac{p_1 A_t}{\dot{m}} = \frac{I_s g_0}{C_F} = \frac{c}{C_F} = \frac{\sqrt{k R T_1}}{k \sqrt{[2/(k+1)]^{(k+1)/(k-1)}}} \quad (3-32)$$

This velocity is only a function of propellant characteristics and combustion chamber properties, independent of nozzle characteristics. Thus, it can be used as a figure of merit when comparing different propellant combinations for combustion chamber performance. The first version of this equation applies in general and allows determination of c^* from experimental values of $\dot{m}$, p_1 , and A_t . The last version gives the ideal

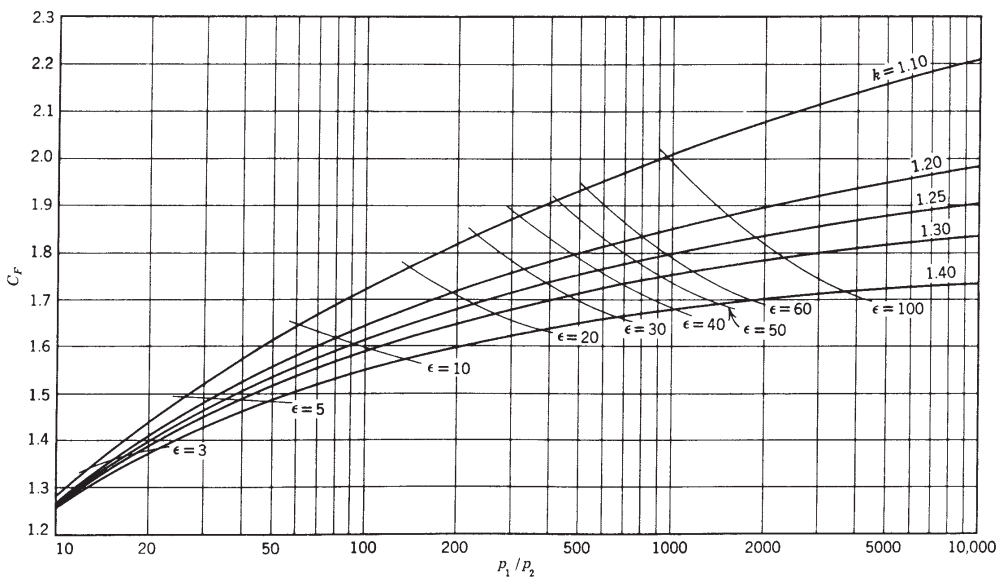


FIGURE 3-5. Thrust coefficient C_F as a function of pressure ratio, nozzle area ratio, and specific heat ratio for optimum expansion conditions ($p_2 = p_3$).

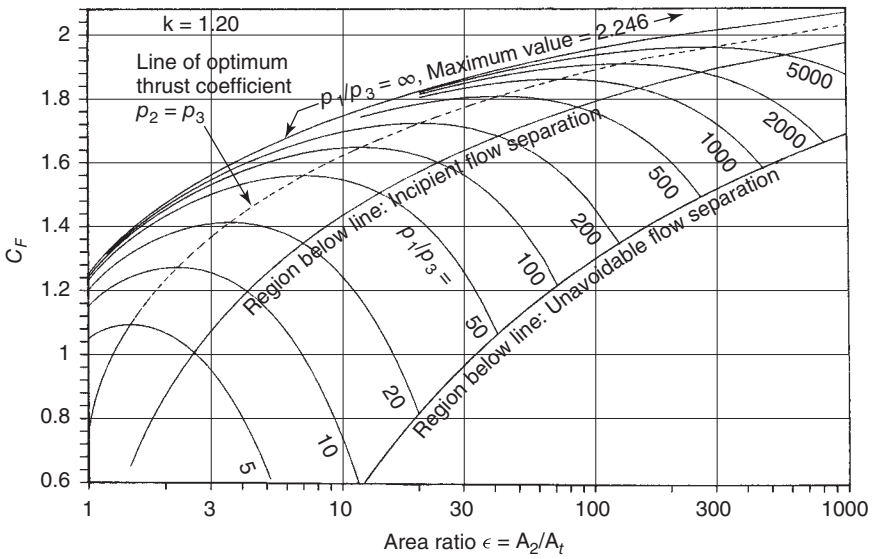


FIGURE 3-6. Thrust coefficient C_F versus nozzle area ratio for $k = 1.20$.

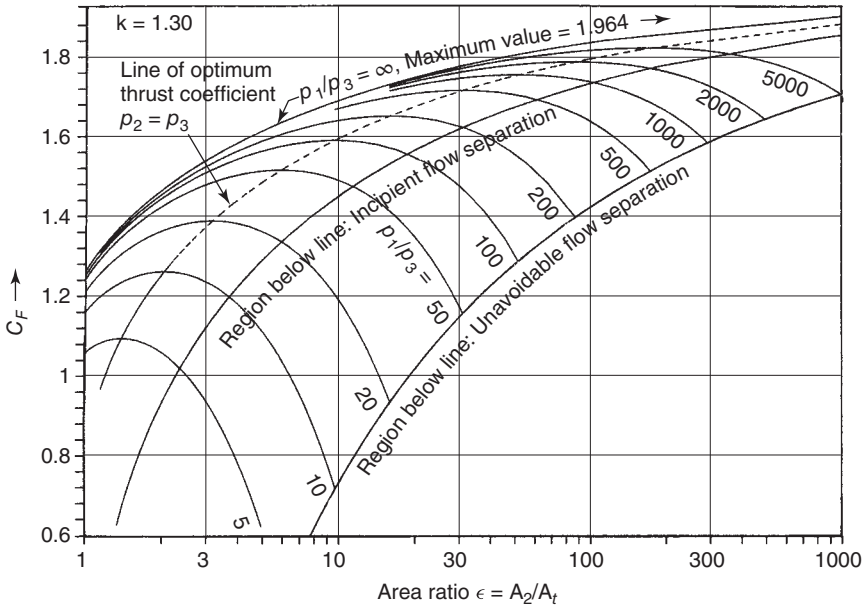


FIGURE 3-7. Thrust coefficient C_F versus nozzle area ratio for $k = 1.30$.

value of c^* as a function of working gas properties, namely k , chamber temperature, and effective molecular mass $\mathfrak{M}$, determined from theory as presented in Chapter 5; some values of c^* are shown in Tables 5-5 and 5-6.

The term c^* -efficiency can be used to express the degree of completion of chemical energy releases in the generation of high-temperature, high-pressure gases in combustion chambers. It is the ratio of the actual value of c^* , as determined from measurements (the first part of Eq. 3-32), to the theoretical value (last part of Eq. 3-32) and has typical values between 92 and 99.5%.

Combining now Eqs. 3-31 and 3-32 allows to express the thrust itself as the mass flow rate ($\dot{m}$) times a function of combustion chamber parameters (c^*) times a function of nozzle expansion parameters (C_F), namely,

$$F = \dot{m} c^* C_F \quad (3-33)$$

Some authors use a term called the *discharge coefficient* C_D , which is merely the reciprocal of c^* . Both C_D and the characteristic exhaust velocity c^* can be used only with chemical rocket propulsion systems.

The influence of *variations in the specific heat ratio* k on various quantities (such as c , $c^* A_2/A_1$, v_2/v_1 , or I_s) is not as large as those from changes in chamber temperature, pressure ratio, or molecular mass. Nevertheless, they are a noticeable factor, as can be seen by examining Figs. 3-2 and 3-4 to 3-7. The value of k is 1.67 for monatomic gases such as helium and argon, 1.4 for cold diatomic gases such as hydrogen, oxygen, and nitrogen, and for triatomic and beyond it varies between 1.1 and 1.3 (methane is 1.11 and ammonia and carbon dioxide 1.33), see Table 7-3. In general, the more complex the molecule the lower the value of k ; this is also true for molecules at high temperatures when their vibrational modes have been activated. Average values of k and $\mathfrak{M}$ for typical rocket exhaust gases with several constituents depend strongly on the composition of the products of combustion (chemical constituents and concentrations), as explained in Chapter 5. Values of k and $\mathfrak{M}$ are given in Tables 5-4, 5-5, and 5-6.

Example 3-4. What is the percentage variation in thrust between sea level and 10 km for a launch vehicle whose rocket propulsion system operates with a chamber pressure of 20 atm and has a nozzle expansion area ratio ϵ of 6? (Use $k = 1.30$.) Also, at what altitude is the performance of this rocket optimum?

SOLUTION. According to Eq. 3-33, the only component of the thrust that depends on the ambient pressure is C_F , the thrust coefficient. This coefficient can be found from Eq. 3-30 or from Fig. 3-7 by following a vertical line corresponding to $A_2/A_1 = 6.0$. We first have to find the ratio p_1/p_3 at sea level and at 10 km. Solving Eq. 3-25 for $p_2/p_1 = 0.0198$, or $p_2 = 0.396$ atm, and using values from Appendix 2 we have

$$\text{Sea level: } p_1/p_3 = 20/1.0 = 20 \quad C_F = 1.33$$

$$10 \text{ km altitude: } p_1/p_3 = 20/0.26151 = 76.5 \quad C_F = 1.56$$

$$\text{Thrust increase} = (1.56 - 1.33)/1.33 = 17.3\%$$

Note that performance at optimum nozzle expansion ($p_2 = p_3 = 0.396 \text{ atm.}$) this nozzle delivers a thrust that is somewhat below the maximum. From Fig. 3–7 we find that for $\epsilon = 6$ the optimum $C_F = 1.52$, which corresponds to 7.2 km elevation, whereas the maximum value is $C_F = 1.63$ in a vacuum.

Under- and Overexpanded Nozzles

An *underexpanded nozzle* discharges the gases at an exit pressure greater than the external pressure because its exit area is too small for an optimum expansion. Gas expansion is therefore incomplete within the nozzle, and further expansion will take place outside of the nozzle exit because the nozzle exit pressure is higher than the local atmospheric pressure.

In an *overexpanded nozzle* the gas exits at lower pressure than the atmosphere as it has a discharge area too large for optimum. The phenomenon of overexpansion inside a supersonic nozzle is indicated schematically in Fig. 3–8, from typical early pressure measurements of superheated steam along the nozzle axis and different back pressures or pressure ratios. Curve AB shows the variation of pressure with

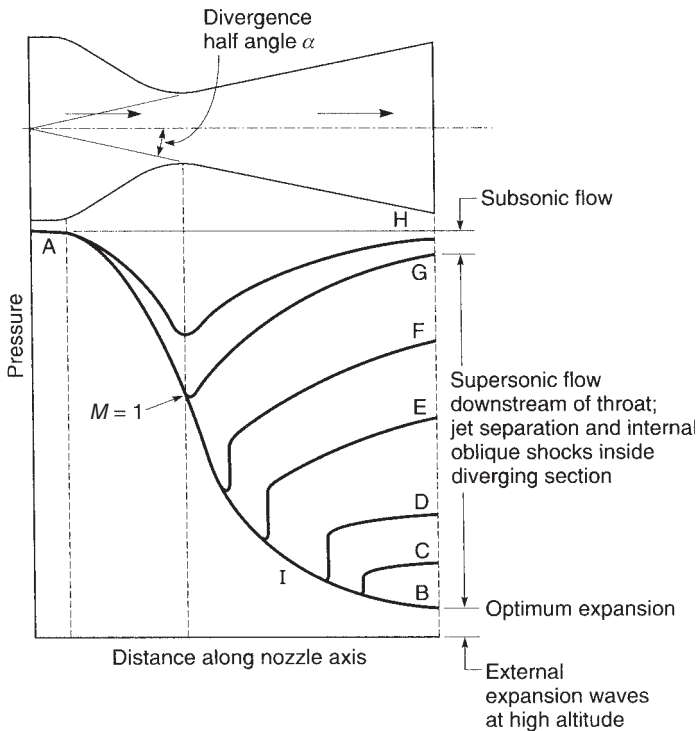


FIGURE 3–8. Distribution of pressures in a converging–diverging nozzle for different flow conditions. Inlet pressure is the same, but exit pressure changes. Based on early experimental data.

optimum back pressure fully corresponding to the nozzle area ratio. Curves AC to AH show pressure variations along the axis for increasingly higher external pressures. The expansion within the nozzle proceeds normally along its initial portion but, for example, at any location after I on curve AD the pressure is lower than the exit pressure and a sudden rise in pressure takes place that is accompanied by *separation* of the flow from the walls (separation is described later in this chapter).

Separation behavior in nozzles is deeply influenced by the presence of compression waves or shock waves inside the diverging nozzle section, which are strong discontinuities and exist only in supersonic flow. The sudden pressure rise in curve ID represents such a compression wave. Expansion waves (below point B), also strictly supersonic phenomena, act to match flows from the nozzle exit to lower ambient pressures. Compression and expansion waves are further described in Chapter 20.

These different possible flow conditions in supersonic nozzles may be stated as follows:

1. When the external pressure p_3 is below the nozzle exit pressure p_2 , the nozzle will flow full but there will form external expansion waves at its exit (i.e., underexpansion). The expansion of the gas inside the nozzle is incomplete and the values of C_F and I_s will be less than at optimum expansion.
2. For external pressures p_3 somewhat higher than the nozzle exit pressure p_2 , the nozzle will continue to flow full. This will occur until p_2 drops to a value between about 10 and 40% of p_3 . The expansion is somewhat inefficient and C_F and I_s will have lower values than an optimum nozzle would have. Weak oblique shock waves will develop outside the nozzle exit section.
3. At higher external pressures, flow separation will begin to take place inside the divergent portion of the nozzle. The diameter of the exiting supersonic jet will be smaller than the nozzle exit diameter (with steady flow, separation remains typically axially symmetric). Figures 3–9 and 3–10 show diagrams for separated flows. The axial location of the separation plane depends on both the local pressure and the wall contour. The separation point travels upstream with increasing external pressure. At the nozzle exit, separated flow remains supersonic in the center portion but is surrounded by an annular-shaped section of subsonic flow. There is a discontinuity at the separation location and the thrust is reduced compared to an optimum expansion nozzle at the available area ratio. Shock waves may exist outside the nozzle in the plume.
4. For nozzles in which the exit pressure is just below the value of the inlet pressure (i.e., curve AH in Fig. 3–8), the pressure ratio is below the critical pressure ratio (as defined by Eq. 3–20) and subsonic flow would prevail throughout the entire nozzle. This condition normally occurs in rocket nozzles for a short time during the start and stop transients.

Methods for estimating pressure at the location of the separation plane inside the diverging section of a supersonic nozzle have traditionally been empirical (see Ref. 3–4). Reference 3–5 describes a variety of nozzles, their behavior, and methods used to estimate the location and pressure at separation. Actual values of pressure for

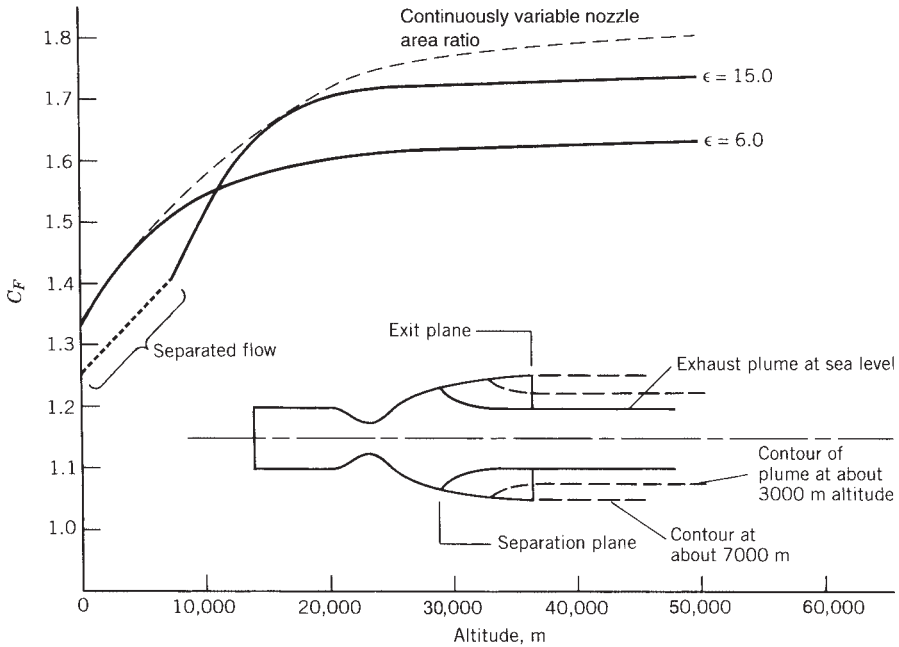


FIGURE 3-9. Thrust coefficient C_F for two nozzles of different area ratios. One has jet separation below about 7000 m altitude. The fully expanded exhaust plume at higher altitudes is not shown in the sketch.

the overexpanded and underexpanded regimes described above are functions of the specific heat ratio and the area ratio (see Ref. 3-1).

The axial thrust direction is not usually altered by separation because steady flows tend to separate uniformly over the divergent portion of a nozzle, but during start and stop transients this separation may not be axially symmetric and may momentarily produce large side forces at the nozzle walls. During normal sea-level transients in large rocket nozzles (before the chamber pressure reaches its full value), some momentary flow oscillations and nonsymmetric separation of the jet may occur while in overexpanded flow operation. The magnitude and direction of these transient side forces may change rapidly and erratically. Because these resulting side forces can be large, they have caused failures of nozzle exit cone structures and thrust vector control gimbal actuators. References 3-5, 3-6, and 3-7 discuss techniques for estimating these side forces.

When flow separates, as it does in a highly overexpanded nozzle, the thrust coefficient C_F can be estimated only when the location of separation in the nozzle is known. The C_F thus determined relates to an equivalent smaller nozzle with an exit area equal to that at the separation plane. The effect of separation is move the effective value of ϵ in Figs. 3-6 and 3-7 to the left of the physical or design value. Thus, with separated gas flows, a nozzle designed for high altitude (large value of ϵ) could have a larger

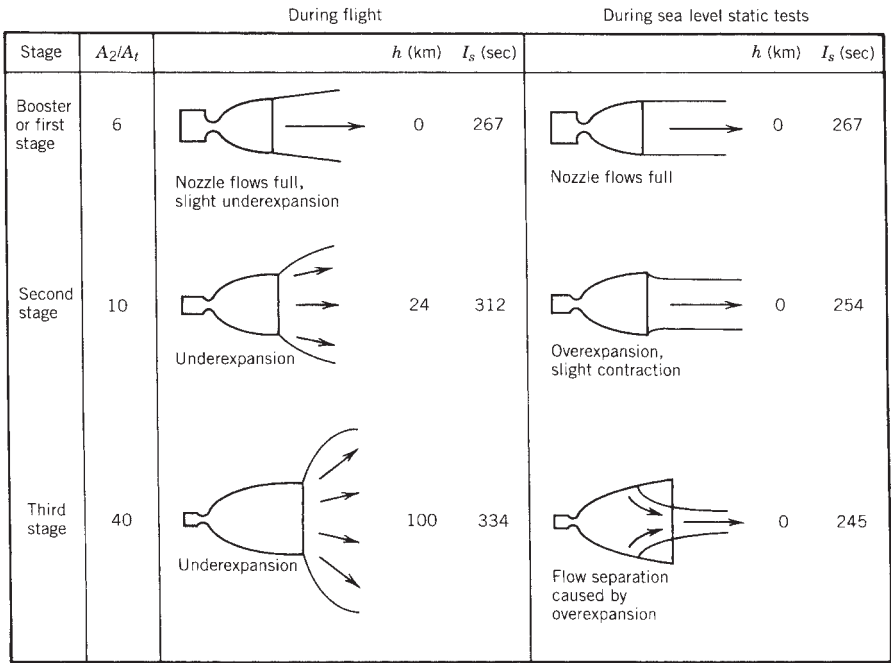


FIGURE 3–10. Sketches of exhaust gas behavior of three typical rocket nozzles of a three-stage launch vehicle. The first vehicle stage has the biggest chamber and the highest thrust but the lowest nozzle exit area ratio, and the top or third stage usually has the lowest thrust but the highest nozzle exit area ratio.

thrust at sea level than expected; in this case, separation may actually be desirable. But with continuous separated flow operation a large and usually heavy portion of the nozzle is not utilized making it bulkier and longer than necessary. Such added engine mass and size decrease flight performance. Designers therefore usually select an area ratio that will not cause separation.

Because of potentially uneven flow separation and its destructive side loads, sea-level static tests of an upper stage or of a space propulsion system with high area ratios are usually avoided because they result in overexpanded nozzle operation; instead, tests are done with a sea-level substitute nozzle (of a much smaller area ratio) often called a “stub nozzle.” However, actual or simulated altitude testing (in an altitude test facility, see Chapter 21) is done with nozzles having the correct area ratio. One solution that avoids separation at low altitudes and has high values of C_F at high altitudes would consist of a nozzle that changes area ratio in flight. This is discussed at the end of this section.

For most atmospheric flights, rocket systems have to operate over a range of altitudes; for a fixed chamber pressure this implies a range of nozzle pressure ratios. The most desirable nozzle for such an application is not necessarily one that gives optimum nozzle gas expansion, but one that gives the best vehicle overall flight

performance (say, total impulse, or range, or payload); this can often be related to a time average over the powered flight trajectory.

Example 3–5. Using the data from Example 3–4 ($p_1 = 20 \text{ atm}$, $k = 1.30$), consider two nozzle area ratios, $\epsilon = 6.0$ and $\epsilon = 15.0$. Compare the performance of these two as a function of altitude by plotting C_F up to 50 km. Assume no shocks inside the nozzle.

SOLUTION. The procedures are the same as those in Example 3–4 and the resulting plot has been shown in Fig. 3–9. Values of C_F can be obtained by following vertical lines for $\epsilon = 6.0$ and 15.0 in Fig. 3–7 corresponding to the different pressure ratios at increasing altitudes. Alternatively, Eq. 3–30 may be used for greater accuracy. The lower area ratio gives better performance at the lower altitudes, but above about 10 km the larger area ratio gives higher values of C_F . Also note that in Fig. 3–9 that the upper dashed line represents what a continuously variable nozzle area ratio could accomplish by matching p_2 to p_3 at all altitudes.

The optimum pressure ratio always occurs when $p_1/p_3 = p_1/p_2$. For the $\epsilon = 15.0$ case, from Fig. 3–5 or 3–7, this ratio is about 180; here $p_3 = 20/180 = 0.111 \text{ atm}$ corresponding to about 15.5 km altitude. Below this altitude such a nozzle is overexpanded. At sea level, where $p_1/p_3 = 20$, separation could occur as indicated in Fig. 3–7. For similarly shaped nozzles, it is estimated that sea-level separation begins to take place at a cross section where the local internal pressure is between 10 and 40% of p_3 or below 0.4 atm. Upon separation, this nozzle would not flow full downstream beyond an area ratio of about 6 or 7 and the gas jet would occupy only the central portion of the nozzle exit (see Figs. 3–9 and 3–10). Weak oblique shock waves and jet contractions would then raise the exhaust jet's pressure to match the value of the surrounding atmosphere. If the jet does not separate, it would reach an exit pressure of 0.111 atm, but this is often an unstable condition. As the vehicle gains altitude, the separation plane would gradually move downstream until, at an altitude of about 7000 m, the exhaust gases could occupy the full exit diverging section.

Figure 3–10 shows behavior comparisons of altitude and sea-level ground-tests of three nozzles and their plumes at different area ratios for a typical three-stage satellite launch vehicle. When fired at sea-level conditions, the nozzle of the third stage with the highest area ratio will experience flow separation and suffer a major performance loss; the second stage will flow full but the external plume will contract; since $p_2 < p_3$, there is a loss in I_s and F . There is no effect on the first stage nozzle. When the second and third stages actually operate at their proper altitude, they would experience no separation.

Example 3–6. A rocket engine test near sea level gives the following data: thrust $F = 53,000 \text{ lbf}$, propellant mass flow rate $\dot{m} = 208 \text{ lbm/sec}$, nozzle exit area ratio $A_2/A_t = 10.0$, actual local atmospheric pressure at test station $p_3 = 13.8 \text{ psia}$, and chamber pressure $p_1 = 620 \text{ psia}$. The test engineer also knows that for the same flow rate (and mixture ratio) the theoretical specific impulse is 289 sec at the standard reference conditions of $p_1 = 1000 \text{ psia}$ and $p_3 = 14.7 \text{ psia}$, and that $k = 1.20$. For many propellants we may assume that the combustion temperature T_1 and k do not vary significantly with chamber pressure. Compare the test performance of this rocket with its equivalent at sea level, standard, and vacuum performance.

SOLUTION. The pressure ratio for the test condition is $620/13.8 = 44.93$; if the test had been conducted at 14.7 psia, this pressure ratio would have been $620/14.7 = 42.18$; for the standard reference conditions the pressure ratio is $1000/14.7 = 68.03$.

Since this nozzle is operating supersonically, we find, from Eqs. 2–5, 3–24, and 3–31, that specific impulse ratios depend only on ratios of the thrust coefficient, that is, $(I_s)_b = (I_s)_a [(C_F)_b / (C_F)_a]$. From Fig. 3–6 or from Eq. 3–30 one obtains $C_F = 1.48$ for the test condition where $\epsilon = 10$, $k = 1.20$, and $p_1/p_3 = 44.93$. The corresponding specific impulse is, from Eq. 2–5, $I_s = 53000/208 = 255$ sec.

The following table shows all results:

	Test Results	Test for Sea Level	Standard Condition	Vacuum Condition
p_1 (psia)	620	620	1000	620
p_3 (psia)	13.8	14.7	14.7	0
p_1/p_3	44.93	42.18	68.03	∞
F (lbf)	53,000	52,480	55,702	60,690
I_s (sec)	255	252.5	268	292
C_F	1.52	1.505	1.60	1.74

Figures 3–9 and 3–10 suggest that a continuous optimum performance for an ascending (e.g., launch) rocket vehicle would require a “rubber-like” diverging section that would lengthen and enlarge so that the nozzle exit area would increase as the ambient pressure is reduced. Such design would then allow the rocket vehicle to attain its best performance at all altitudes as it ascends. As yet we have not achieved such flexible, stretchable mechanical nozzle designs with full altitude compensation similar to “stretching rubber.” However, there are a number of practical nozzle configurations that can be used to alter nozzle shape with altitude and obtain peak performance at different altitudes. These are discussed in the next section.

Influence of Chamber Geometry

When the chamber has a cross section that is about four times larger than the throat area ($A_1/A_t > 4$), the chamber velocity v_1 , may be neglected, as was mentioned in introducing Eqs. 3–15 and 3–16. However, some vehicle space- or weight-constraints for liquid propellant engines often require smaller thrust chamber areas, and for solid propellant motors grain design considerations may lead to small void volumes or small perforations or port areas. In these cases, v_1 ’s contribution to performance can no longer be neglected. Gases in the chamber expand as heat is being added. The energy expended in accelerating these expanding gases within the chamber will also cause a pressure drop, and processes in the chamber can be adiabatic (no heat transfer) but not isentropic. Losses are a maximum when the chamber diameter equals the nozzle throat diameter, which means that there is no converging nozzle section. This condition is called a *throatless rocket motor* and is found in a few tactical missile booster applications, where there is a premium on minimum inert mass and length. Here, flight performance improvements due to inert mass savings have supposedly outweighed the nozzle performance loss of a throatless motor.

Because of the significant pressure drops within narrow chambers, chamber pressures are lower at the nozzle entrance than they would be with a larger A_1/A_t .

TABLE 3–2. Estimated Losses for Small-Diameter Chambers

Chamber-to-Throat Area Ratio	Throat Pressure (%)	Thrust Reduction (%)	Specific Impulse Reduction (%)
∞	100	0	0
3.5	99	1.5	0.31
2.0	96	5.0	0.55
1.0	81	19.5	1.34

$k = 1.20$; $p_1/p_2 = 1000$.

This causes a small loss in thrust and specific impulse. The theory of such losses is given in the Second and Third Editions of this book, and some results are listed in Table 3–2.

3.4. NOZZLE CONFIGURATIONS

Several different proven nozzle configurations are available today. This section describes their geometries and performance. Other chapters in this book (6, 8, 12, 15, and 18) discuss nozzle materials, heat transfer, and their applications with mention of certain requirements, design, construction, and thrust vector control. Nozzles and chambers are typically circular in cross section and must have a converging section, a throat or minimum cross section, and a diverging section. Nozzles are apparent in Figs. 1–4, 1–5, 1–8, 2–1, 3–10, as well as in Figs. 3–11, 3–13, 11–1, 12–1 to 12–3, and 15–6 and 15–8. Refs. 3–4 and 3–8 describe several other nozzle configurations.

The *converging nozzle section* shape between the chamber and the throat does not significantly affect nozzle performance. Its subsonic flow can easily be turned with very low pressure drops and almost any given radius, cone angle, wall contour curve, or nozzle inlet shape is satisfactory. A few small attitude control thrust chambers have had their nozzle turn at 90° from the combustion chamber axis without measurable performance loss. The *throat contour* itself is also not very critical to performance, and any smooth curved shape is usually acceptable. Since pressure gradients are high in these two regions, the flow will always adhere to the walls. Principal differences in nozzle configurations are found in the diverging supersonic-flow section, as described below. In general, internal wall surfaces throughout the nozzle should be smooth and reflective to minimize friction, radiation absorption, and convective heat transfer, which is enhanced by surface roughness. Gaps, holes, sharp edges, and/or protrusions must be avoided.

Six different nozzle configuration types have been considered and they are depicted in Fig. 3–11. The first three sketches show the more common conical and bell-shaped nozzles and they are further described in this chapter. The last three have a center body inside the nozzle in order to provide altitude compensation (this means that outside the nozzle the hot-gas boundaries can grow as the outside pressure decreases allowing a more optimum expansion with altitude). Although these last three have been thoroughly ground tested, to date none of them has flown in a

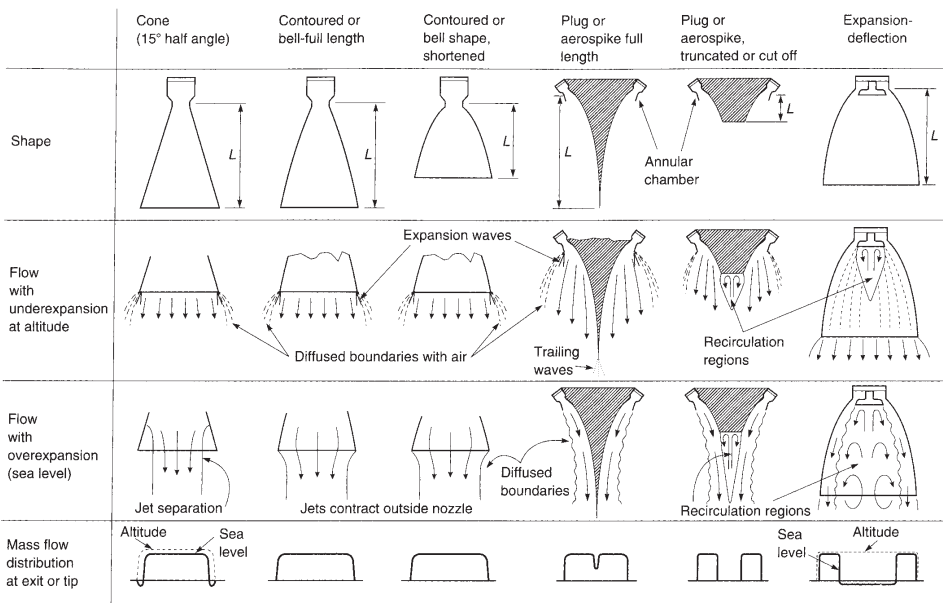


FIGURE 3–11. Simplified diagrams of different generic nozzle configurations and their flow effects.

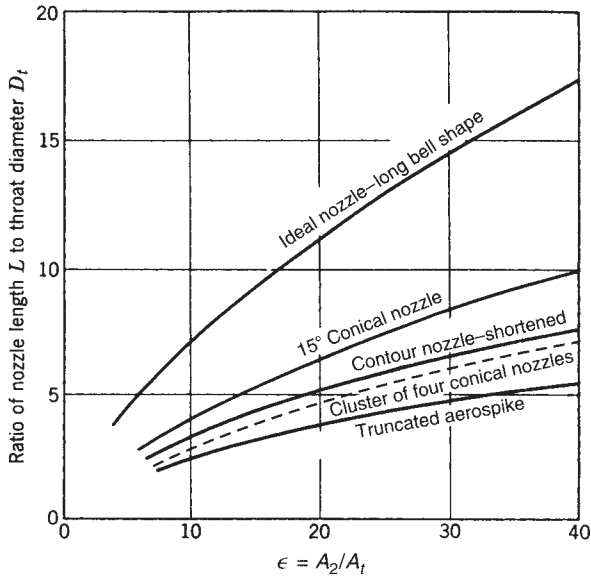


FIGURE 3–12. Length comparison of several types of nozzles. Taken in part from Ref. 3–9.

production space launch vehicle (see Eighth edition). The lengths of several nozzle types are compared in Fig. 3–12. The objectives for a good nozzle configuration are to obtain the highest practical I_s minimizing inert nozzle mass and conserving length (shorter nozzles can reduce vehicle length, vehicle structure, and vehicle inert mass).

Cone- and Bell-Shaped Nozzles

The *conical nozzle* is the oldest and perhaps the simplest configuration. It is relatively easy to fabricate and is still used today in many small nozzles. A theoretical correction factor λ must be applied to the nozzle exit momentum in any ideal rocket propulsion system using a conical nozzle. This factor is the ratio between the momentum of the gases exhausting with a finite nozzle angle 2α and the momentum of an ideal nozzle with all gases flowing in the axial direction:

$$\lambda = \frac{1}{2} (1 + \cos \alpha) \quad (3-34)$$

For ideal rockets $\lambda = 1.0$. For a rocket nozzle with a divergence cone angle of 30° (half angle $\alpha = 15^\circ$), the exit momentum and therefore the axial exhaust velocity will be 98.3% of the velocity calculated by Eq. 3–15b. Note that the correction factor λ only applies to the first term (momentum thrust) in Eqs. 2–13, 3–29, and 3–30 and not to the second term (pressure thrust).

A half angle value of 15° has become the unofficial standard of reference for comparing correction factors or energy losses or lengths from different diverging conical nozzle contours.